# Graphic Design Studio 3

## Brief 2

## A COPY OF A COPY

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# Project Overview

The core of this project is to create the classroom through a "model product".

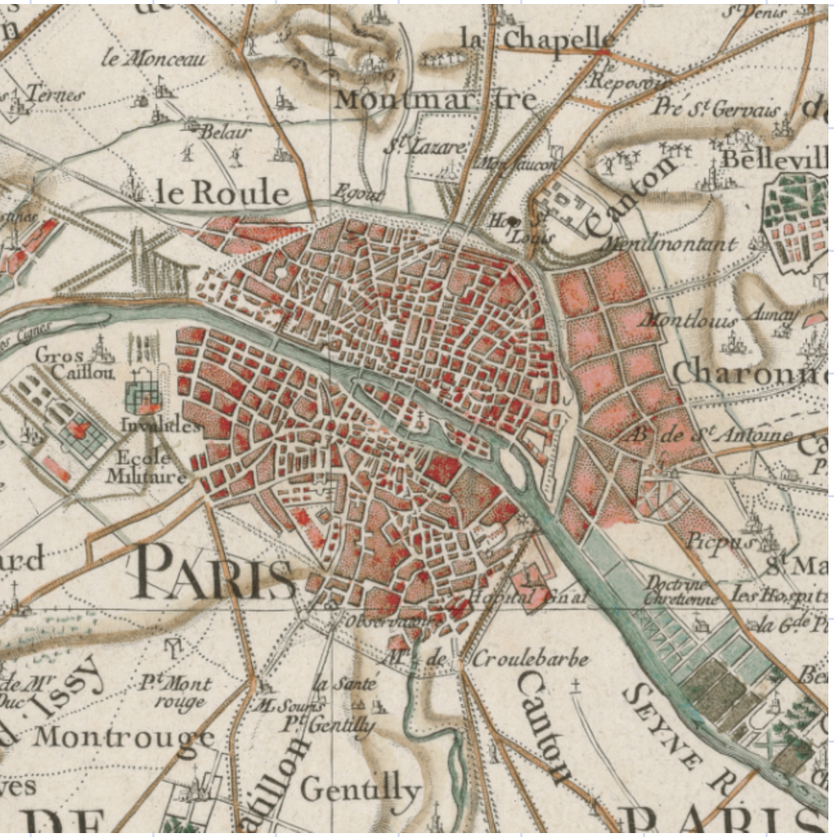
The inspiration for the project came from the fact that when the Brief title was first mentioned, and the classroom was selecting objects, ideas, and brands related to "a Copy of a Copy", I chose Carte de Cassini (A very accurate map of France drawn in the 18th century) and looked around, all of which were classrooms: the classroom itself was a product of extracting geography from a map, designing buildings, and then "copying" them into reality.

The implementation of this project comes from a combination of my personal interests and skills, as well as the tools I can use. I have no experience in creating real-life building models, which gave me an idea. I kept trying and making mistakes, and finally found a "greatest common divisor". Using this method, I modelled this part of the classroom with materials.

# Research - Related Topic

Carte de Cassini, or the Cassini map, is a map of France drawn by the Cassini family over four generations in the 18th century. The map scale is 1 line to 100 toises, which is a 1/86400 scale.

The map is drawn based on a large number of geographic triangulations, and it is also the first map based on geographic triangulation, which can achieve astonishing accuracy in an era without satellite images.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le Monceau
Ternes
Montmartre
Belair
St Lazare
Pré St Gervais
Belleville
le Roule
Egoust
Canton
Menilmontant
Montlouis
Charonne
Gros Caillou
Invalides
Ecole Militaire
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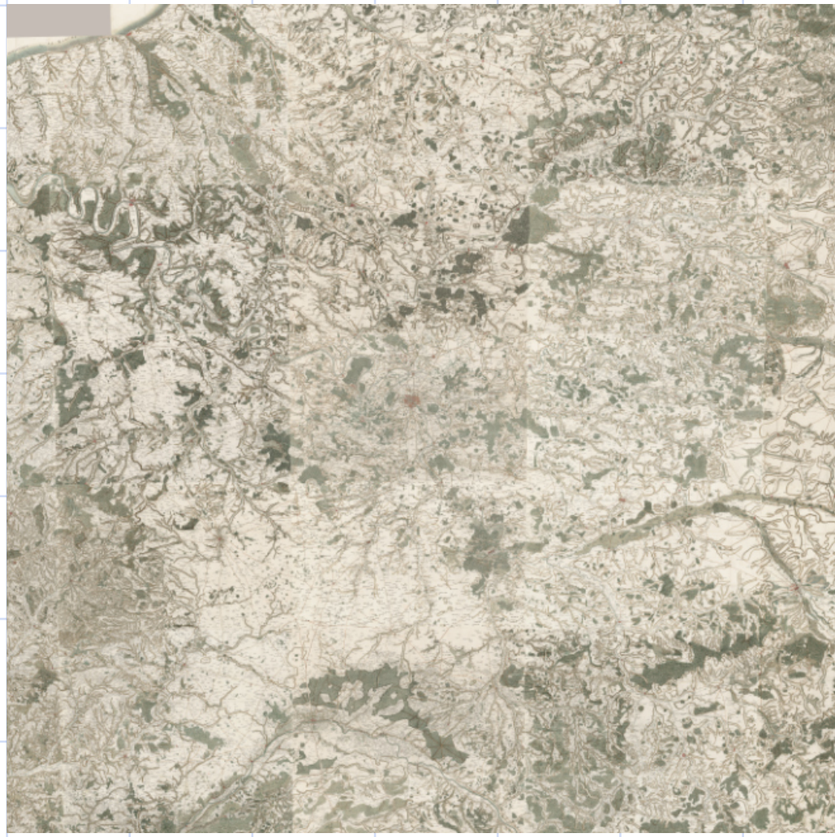

# Research - Related Topic

The LEGO manual uses step-by-step guidance to directly display the construction process, relying entirely on images, and breaks down the process into a series of simple and understandable small tasks.

Not rely on language for information transmission; even Children who can't read can perform every step of the operation by observing pictures.

The LEGO manual also displays a complete parts list, and each step includes the required part labels. This allows users to quickly identify and prepare the required materials.

Ages/edades
6+
40276

6223084 / 6223178

Warning! Choking hazard. Small parts.
Achtung! Erstickungsgefahr. Kleine Teile.
Attention ! Danger d'étouffement. Petits éléments.
Avvertenza! Rischio di soffocamento. Piccole parti.
Waarschuwing! Verstikkingsgevaar. Kleine onderdelen.
¡Advertencia! Peligro de atragantamiento. Partes pequeñas.
Advarsel! Kvælningsfare. Små dele.
Viðvörun! Köfnunarhætta. Litlir hlutir.
Varoitus! Tukehtumisvaara. Pieniä osia.
Varning! Kvävningsrisk. Små delar.
Advarsel! Kvelningsfare. Små deler.
Atenção! Risco de asfixia. Pequenas partes.
Προειδοποίηση! Κινδυνος πνιγμού. Μικρά μέρη.
警告!小部品があります。誤飲・窒息の危険があります。
警告！小心窒息，当心小物件。
경고! 질식 위험. 작은 부품 포함.
Внимание! Риск удушения. Мелкие детали.
Ostrzeżenie! Niebezpieczeństwo udławienia się. Małe części.
Upozornění! Nebezpečí zaknutí. Malé části.
Upozornenie! Nebezpečenstvo dusenia. Malé časti.
Figyelmeztetés! Fulladasveszély. Kis alkatrészek.
Попередження! Ризик задухи. Дрібні деталі.
Upozorenje! Opasnost od gušenja. Mali dijelovi.
Упозорење! Опасност од гушења. Мали делови.
Предупреждение! Опасност од задушување. Мали делови.
Opozorilo! Nevarnost zadušitve zaradi tujka. Majhni deli.
Avertisment! Pericol de sufocare internă. Părți mici.
Внимание! Опасност от задавяне. Малки части.
Bridinājums! Aizrīšanās risks. Sīkas detaļas.
Hoiatus! Kägistamisoht. Väikesed osad.
Įspėjimas! Pavojus užspringti. Smulkios detalės.
Uyarı! Boğulma Tehlikesi (Havasızlıktan). Küçük Parçalar.
تحذير! خطر الاختناق. أجزاء صغيرة
Peringatan! Bahaya tersedak. Bagian kecil.
Amaran! Bahaya tercekik. Bahagian kecil.

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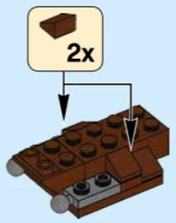


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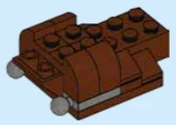

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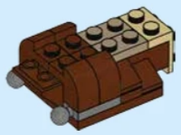

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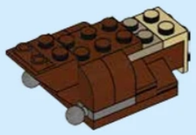

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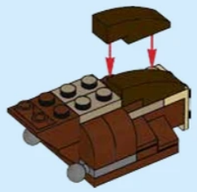

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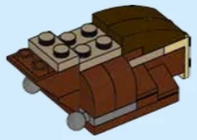

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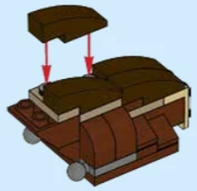

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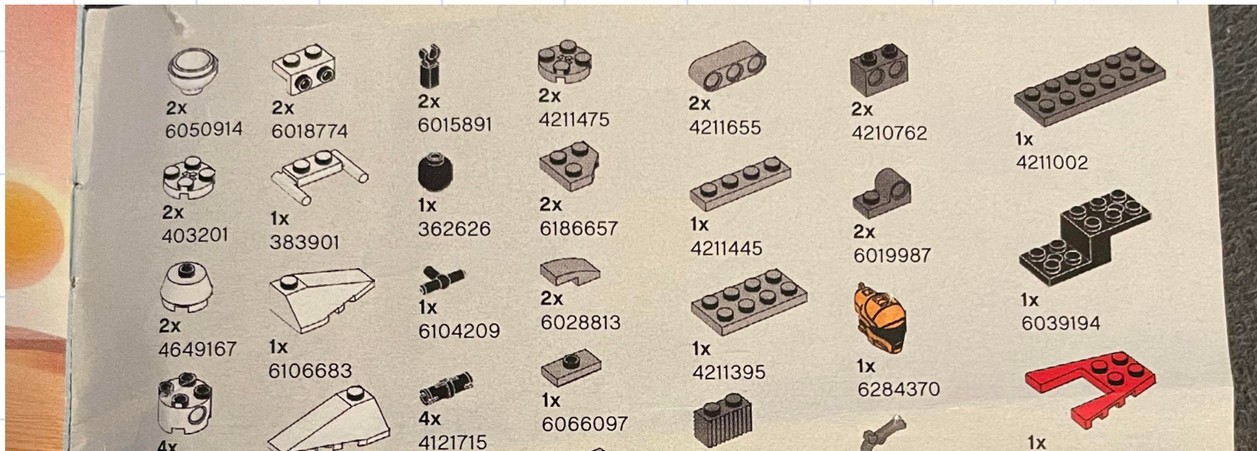
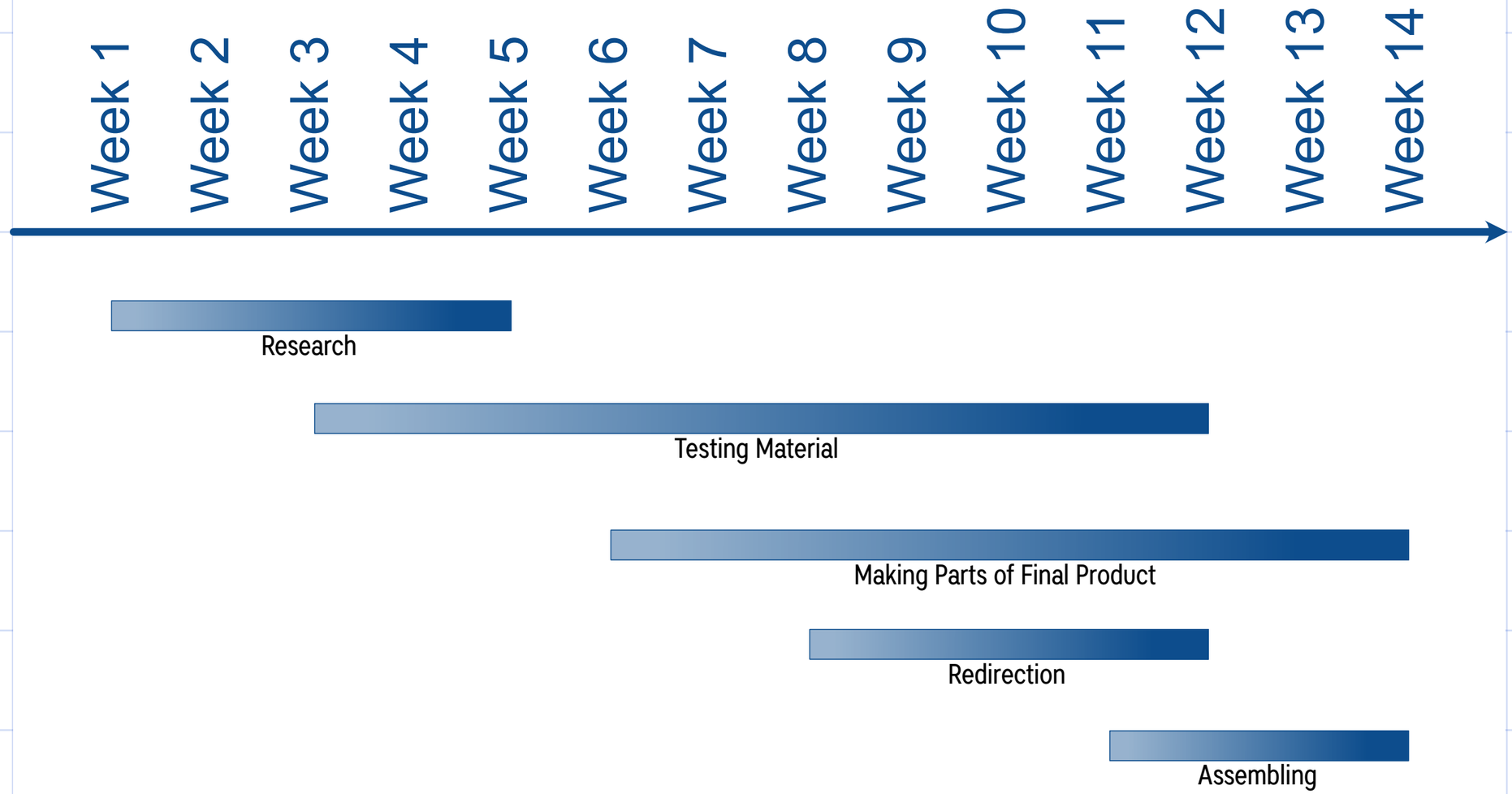


# Project Direction

I see the classroom I am in as an architect's architectural floor plan that have been copied to reality, I will copy this copy back to a design plan, and copy it to reality through a model to comply with the "A COPY OF A COPY" title.

# Timeline


Week 1
Week 2
Week 3
Week 4
Week 5
Week 6
Week 7
Week 8
Week 9
Week 10
Week 11
Week 12
Week 13
Week 14
Research
Testing Material
Making Parts of Final Product
Redirection
Assembling

# Timeline

This is a big project.

How do I make the model?

How do I determine the proportion?

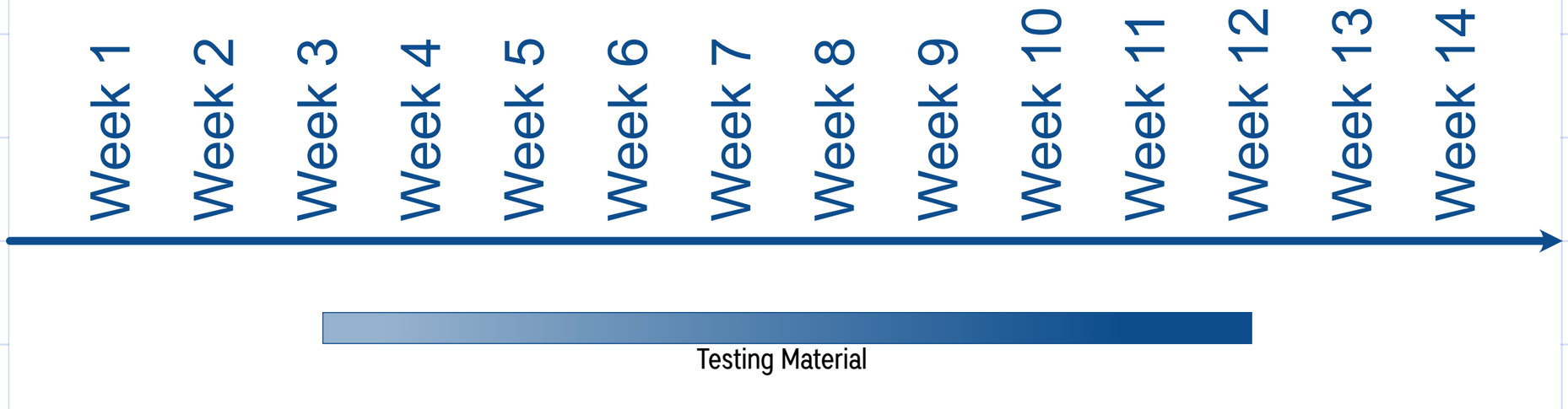
How do I showcase the results?

Where did others encounter challenges when making models?

Which parts require more time?

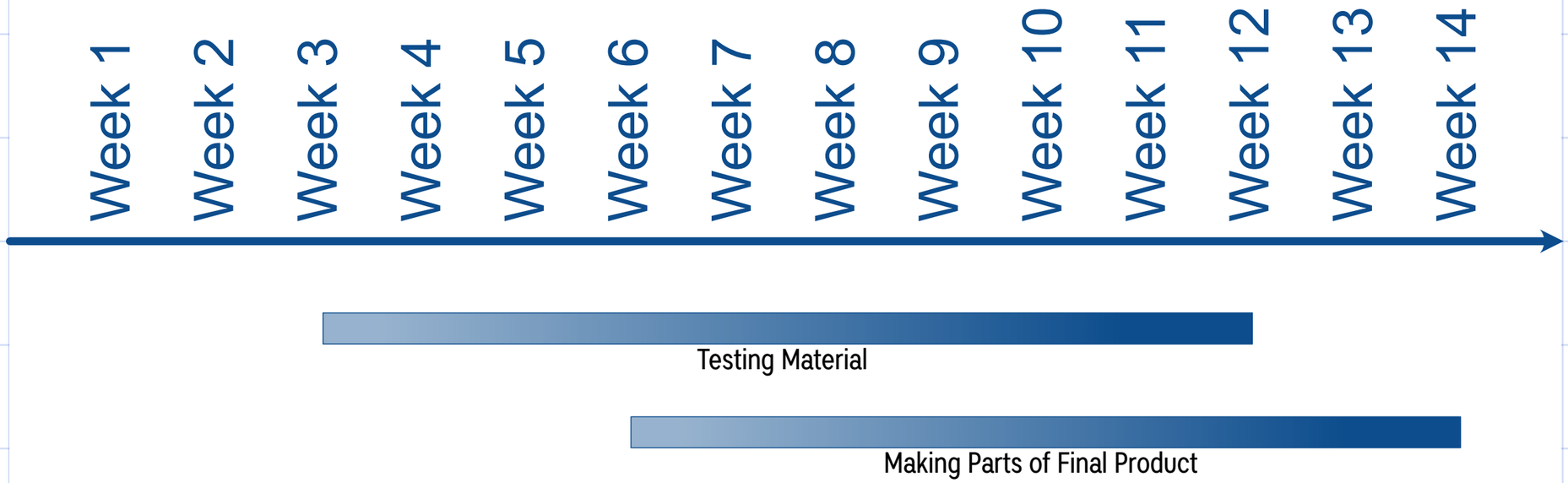
What do I need to learn to achieve? etc...

# Timeline




I don't want to imitate a classroom just by using "similar colours" and then finishing the work. I want similar textures and materials without spending so much time on production. Find the optimal balance between "similarity" and "ease", and implement it.

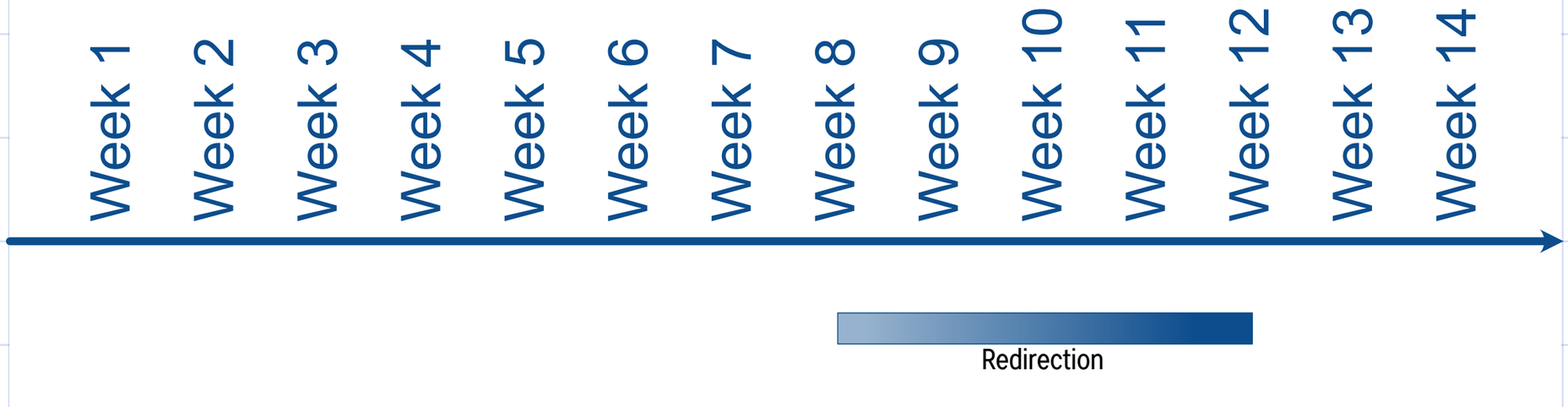
# Timeline




The making of final parts and testing materials' process are somewhat synchronised.

During the making process, I repeatedly found parts that I thought would be used for the final presentation, but later found better material/technique replacements and replaced all appearances (physically, design-wise, and in the manual display).

# Timeline




In the process of materialising the model, to further demonstrate the relationship between this project and Graphic Design, boxes and manuals were introduced, which greatly increased the workload, due to manuals and boxes not originally envisioned in the project direction, many topics and designs needed to be reworked and conceptualised, or even thought about and brainstormed for the first time.

# Timeline

Assemble all the final results together to create the final model, and render all the 3D models into an Exploded Assembly View in the manual.

# Research - Project Questions

What scale should I go for?

How to package the object, what to include?

What will I need to "copy" many classroom materials and textures:

- What material do I need to prepare?
- What tools and machines do I need to use?
- What tools and machines might I need to learn?

And many more questions appear mid-process...

# Project Overview

~~What scale should I go for?~~

My model's EYE placement is tentatively set at the mantelpiece, so the final model's area taken up should not be larger than A4 paper. When the project was halfway through, I found out I also need to consider the box, and the box's area taken up would not be larger than A3 paper.

Based on the length, width, and height of the room, I chose a 1:40 ratio, which is also one of the common ratios for Interior Architectural Scale models. This allows me to refer to the production of some Interior Architectural Scale models

# Project Overview

~~How to package the object, what to include?~~

The model itself, the box that contains parts and manuals.

Therefore when displaying, there should mostly be a fully constructed model and a box, which includes all the components and manuals used for assembly.

# Project Overview

~~What will I need to "copy" many classroom materials and textures:~~

There will be a whole section about this; the direction is to focus on:

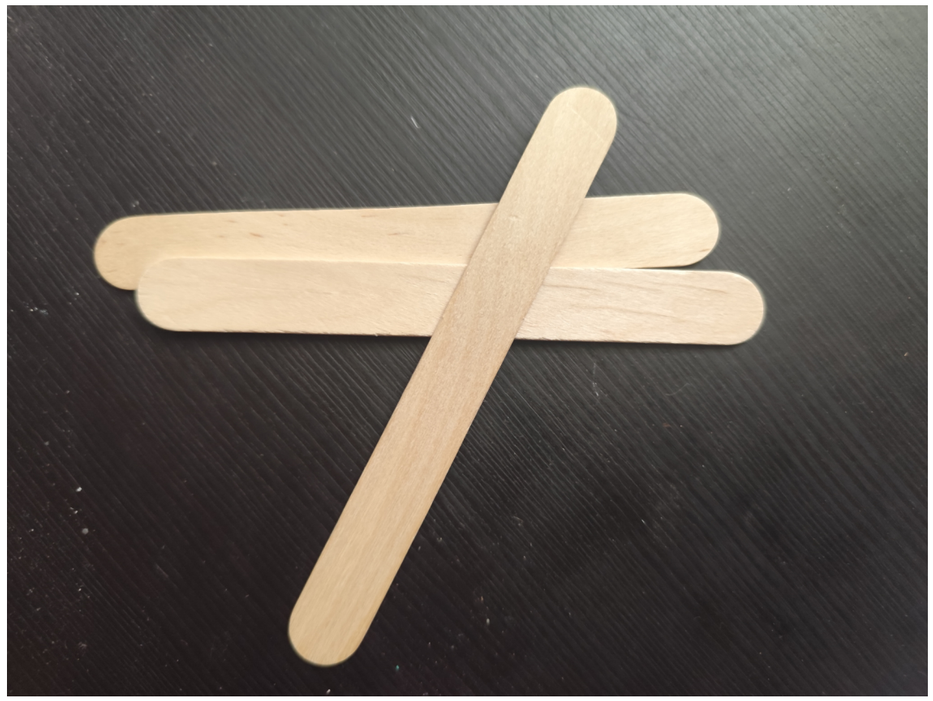
- Architectural Materiality - The feeling, the texture of the material
- Material Representation in Architectural Models - Simulate the original material's texture being scaled down (eg., use sanded paint to replace concrete)
- Material Translation Architecture - Use another material to simulate the original material that preserve feature of original material.

# Testing Material

Structurally, the classroom consists of four major elements:

- Wooden Flooring
- Concrete Walls with White Paint
- Windows
- Greyish-green Framework

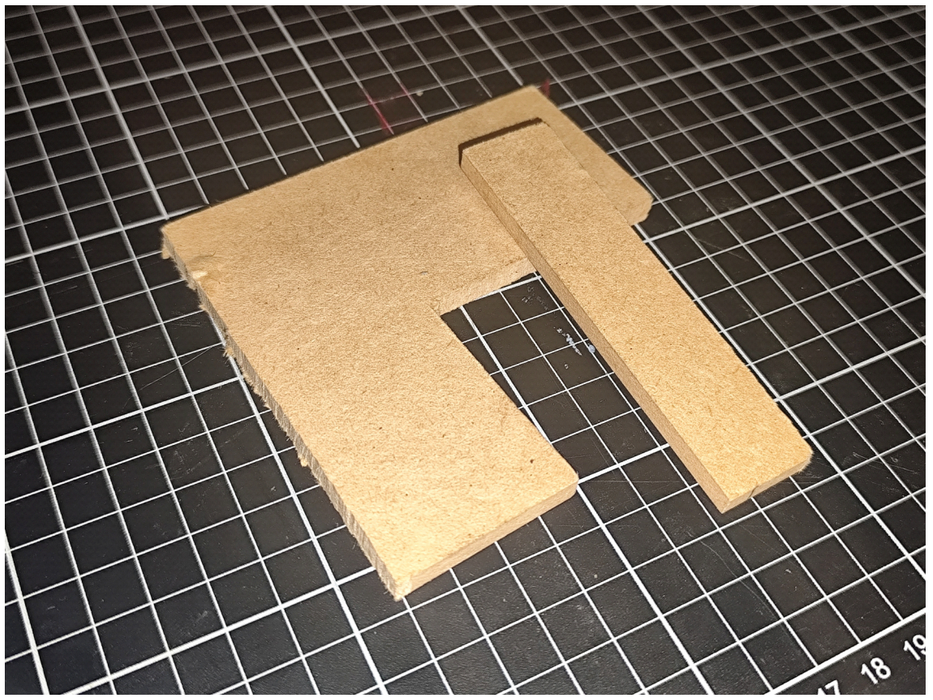
# Testing Material - Wooden Floor



## Ice Cream Stick

- - Shatters the moment I try to saw it

# Testing Material - Wooden Floor



## Sawed Compressed Wood Plank

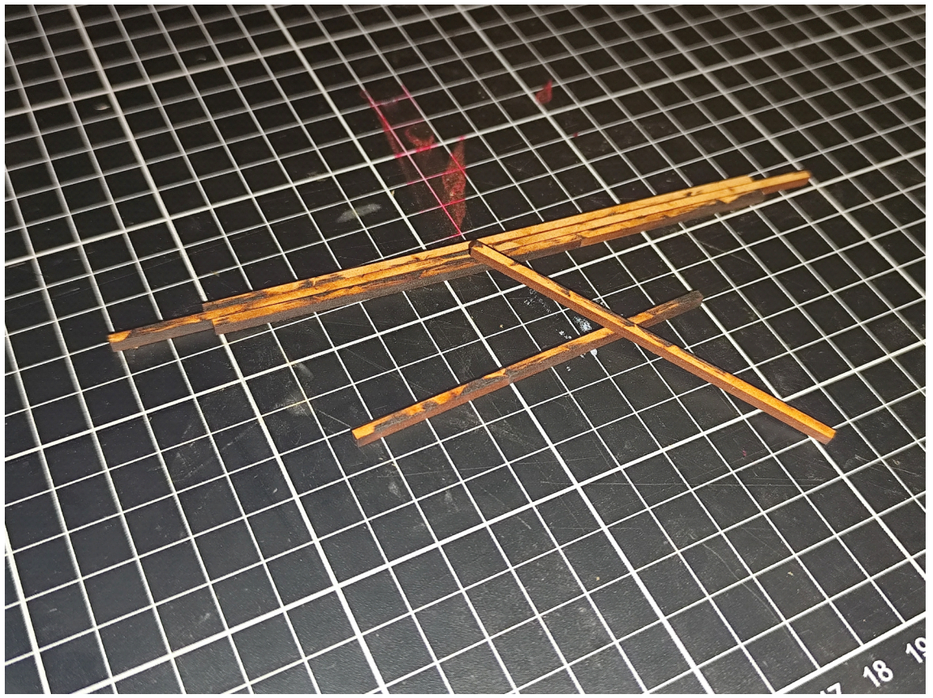
- At a scale of 1:40, the width of a single plank is approximately 2mm, making it difficult to control the width, and deviations exceed this number.

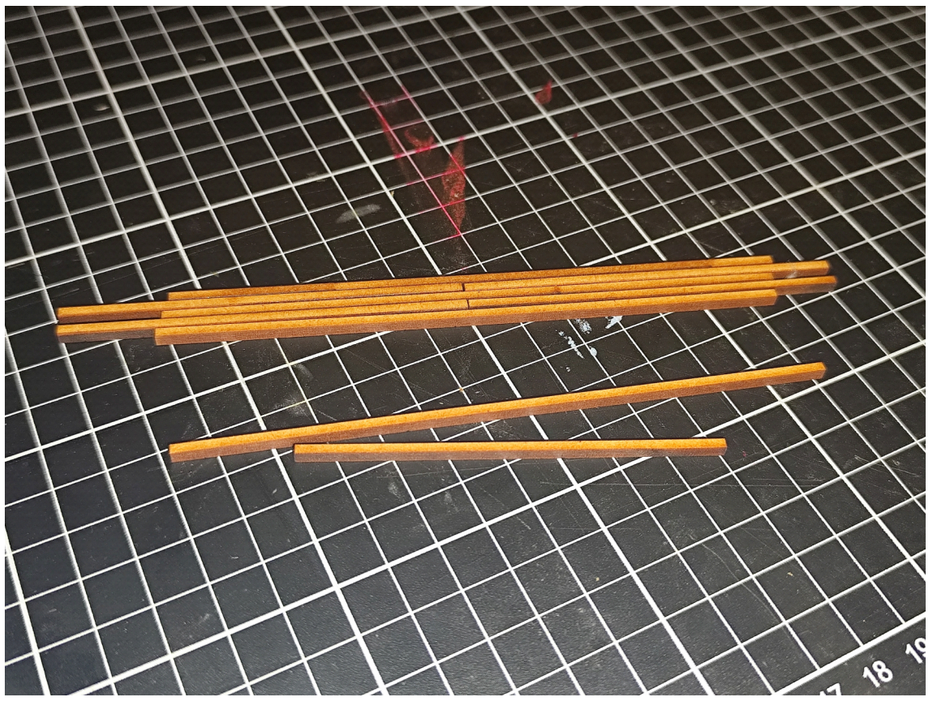
- Cannot be too thin; otherwise it will shatter or delaminate during sawing.

- Extremely slow to saw; even if no errors occur, it takes more than a day to cut out hundreds of wooden strips to simulate the planks used for wood floors

- Sawing out requires processing of rough and sharp edges and corners.

# Testing Material - Wooden Floor





Laser-Cut MDF Board (Cut Through)

+ Compared to sawn compressed wood plank, very fast cutting.

+ Can be batch-produced through grid cutting.

+ Very precise.

+ Compared to Sawed Compressed Wood Plank, very thin (3mm).

+ The corners are burnt, no need further processing.

# Testing Material - Wooden Floor

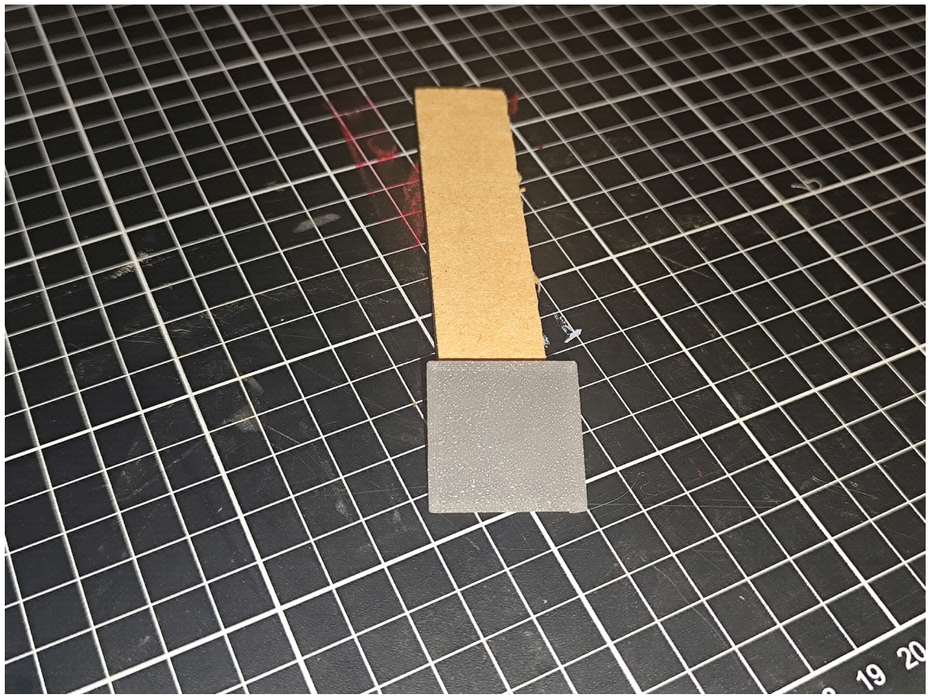
## Laser-Cut MDF Board (Etching)

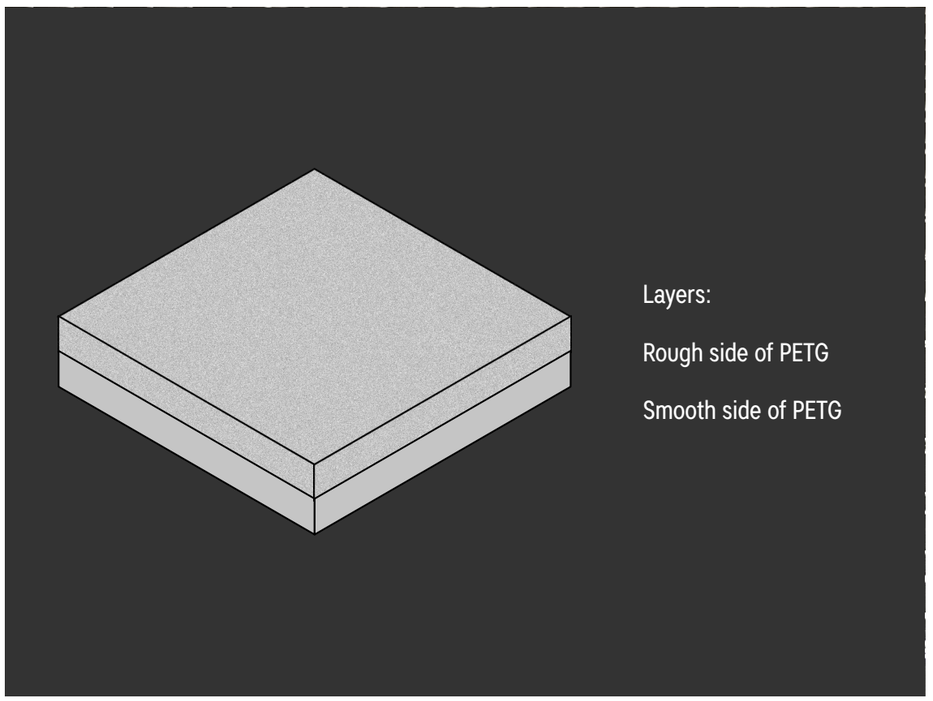
+ Inherits most of the benefits of Laser Cut MDF Board (Cut Through); it does not cut through the MDF Board, which eliminates the process of assembling hundreds of small wooden strips.

+ The floor of the model no longer requires an additional layer as its base.

- The etched border is not as visible compared to cutting it through

# Testing Material - Wall



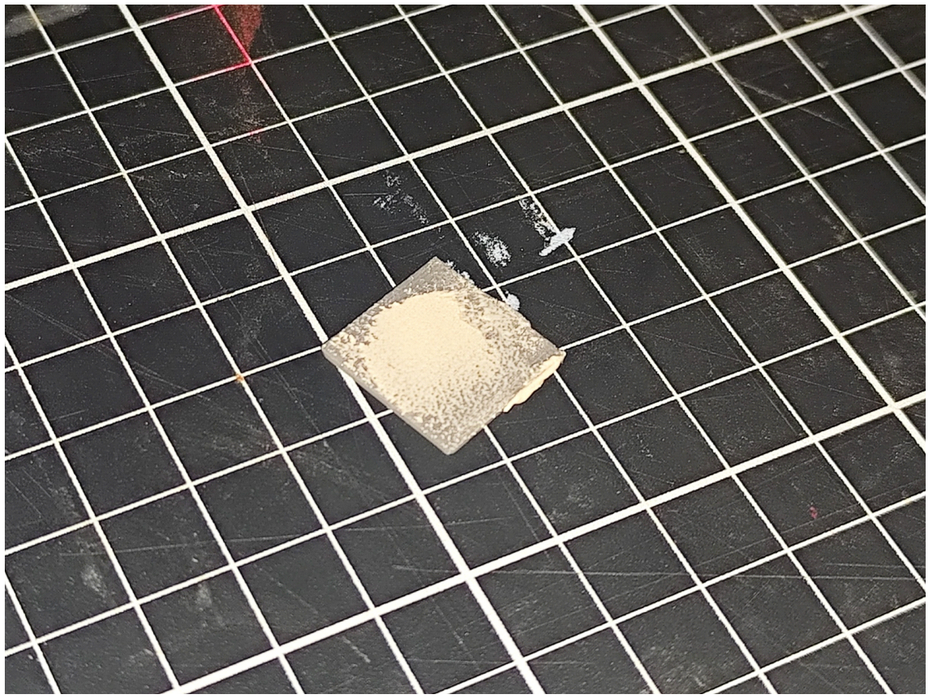



3D Printed PETG on Rough Texture Printing Plate

Not a choice considered; just noticed its features during testing

- Hard and plastic texture
- Uneven surface, but too slippery
+ Strong ductility; can bend to an extent
- Need a specific colour of filament

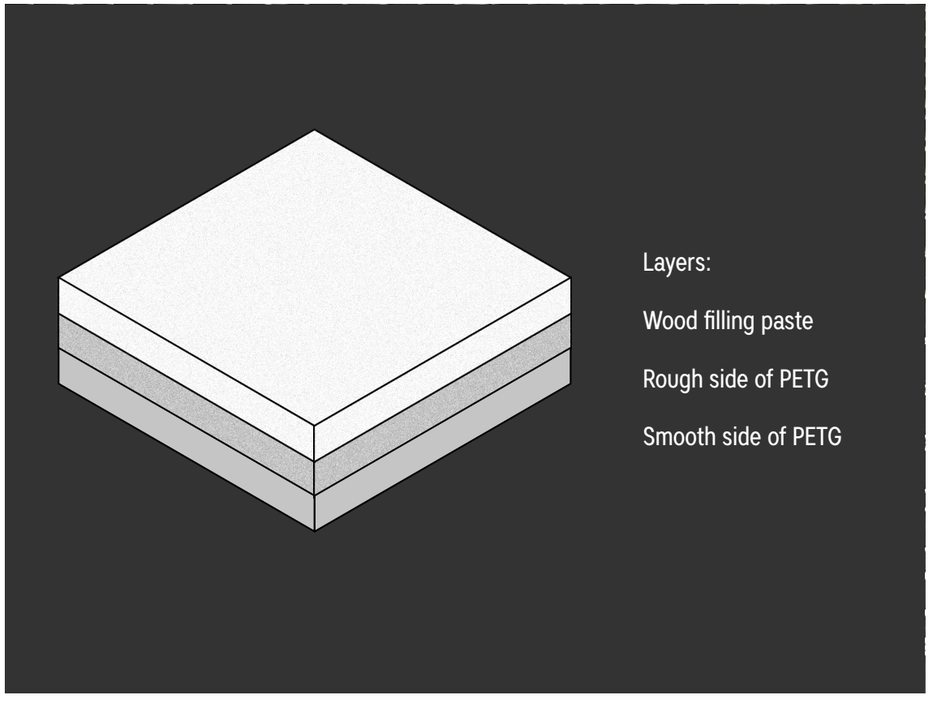
# Testing Material - Wall



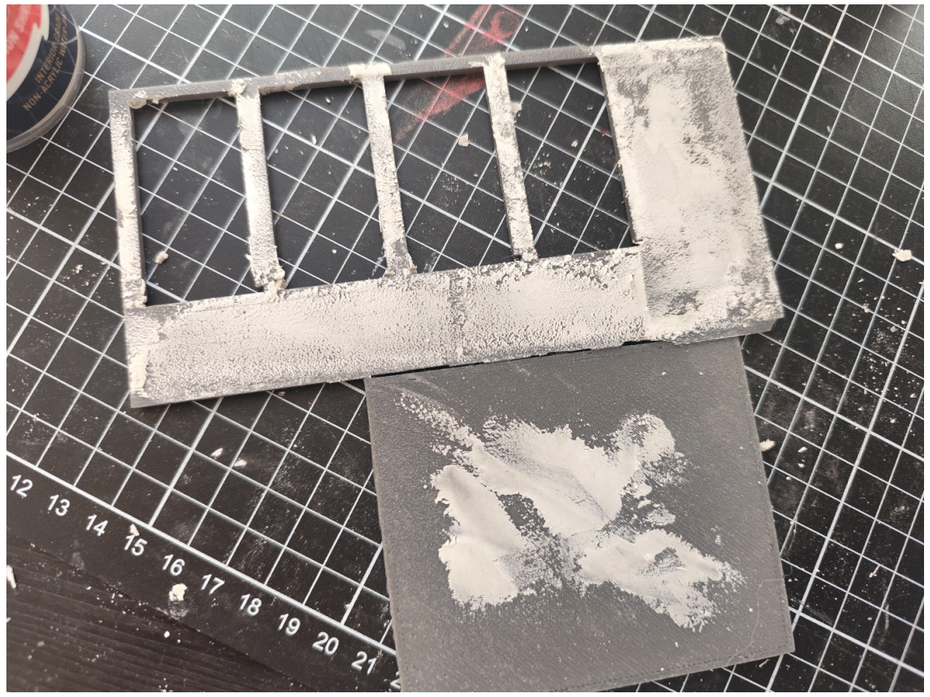
3D-Printed PETG on Rough Texture Printing Plate, coated with Wood Filling Paste

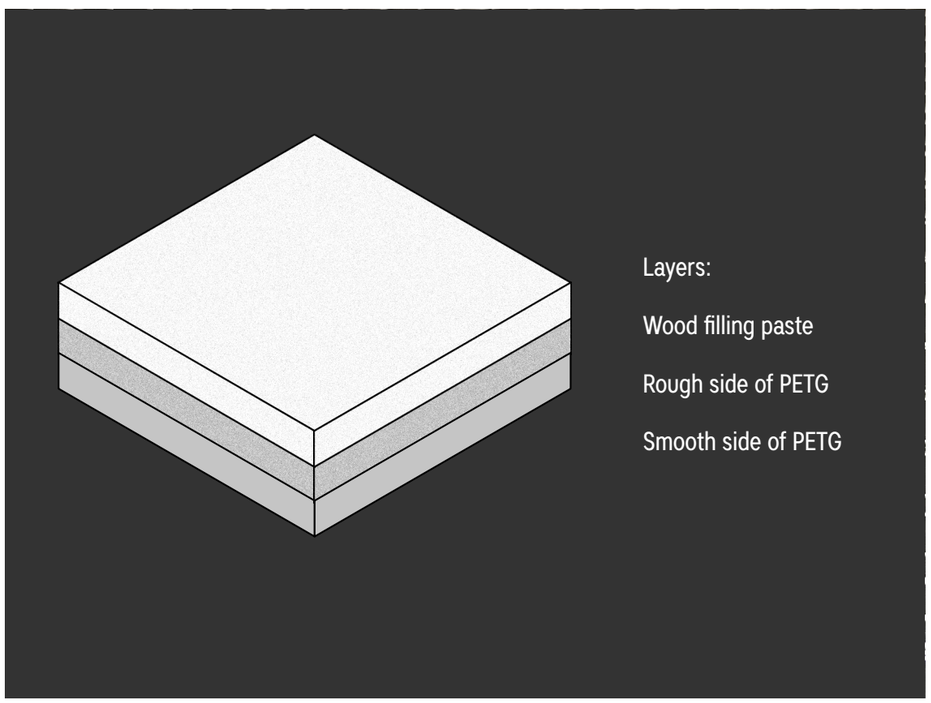
+ Feels like a scaled-down version of concrete texture, rough and flat

+ Colours close to the original wall to choose from



# Testing Material - Wall





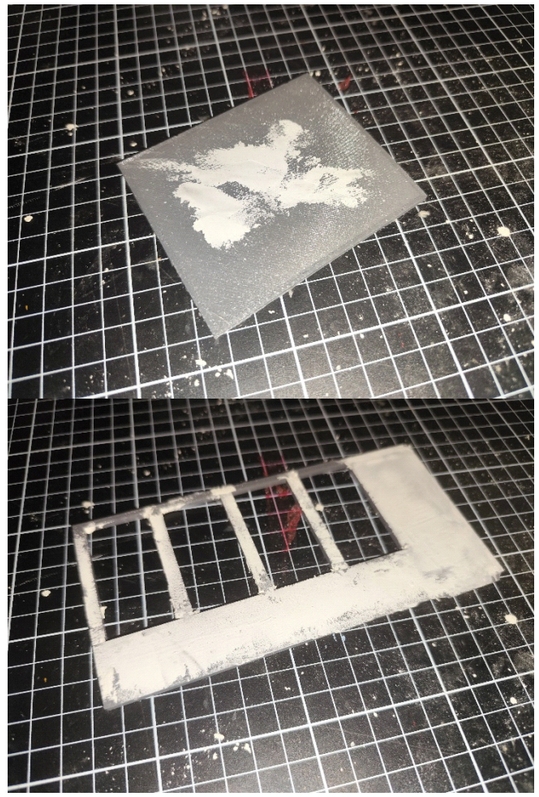

3D Printed PETG on Rough Texture Printing Plate, coated with Wood Filling Paste

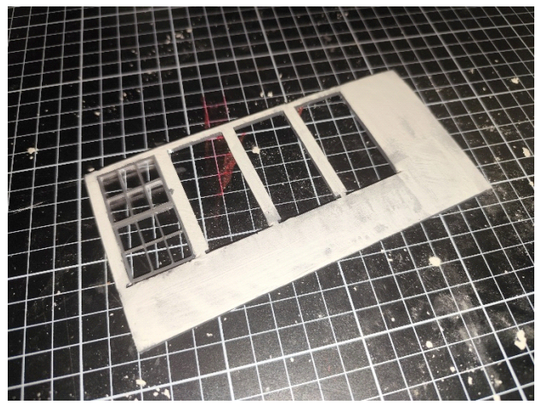
But later discovered that this coating also has these features:

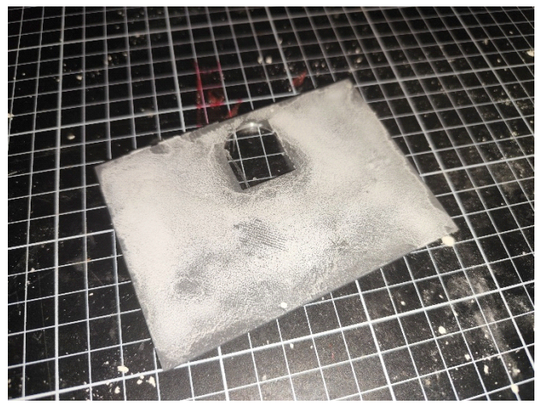
- Rubber putty texture, very difficult to apply in a uniform thickness

- Slightly sticky, difficult to apply in areas with holes and openings

# Testing Material - Wall







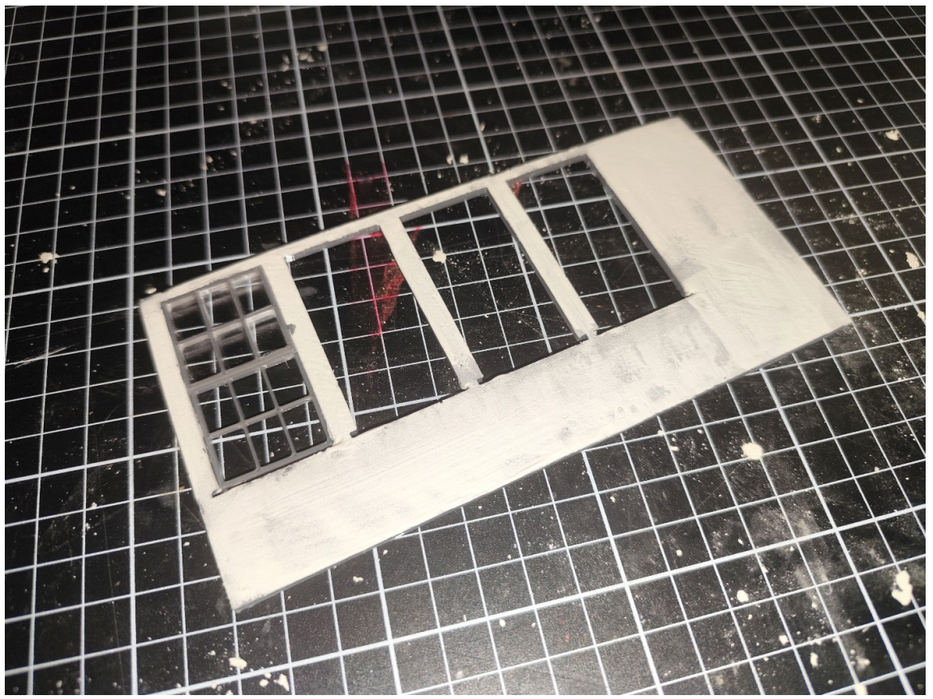
Dry

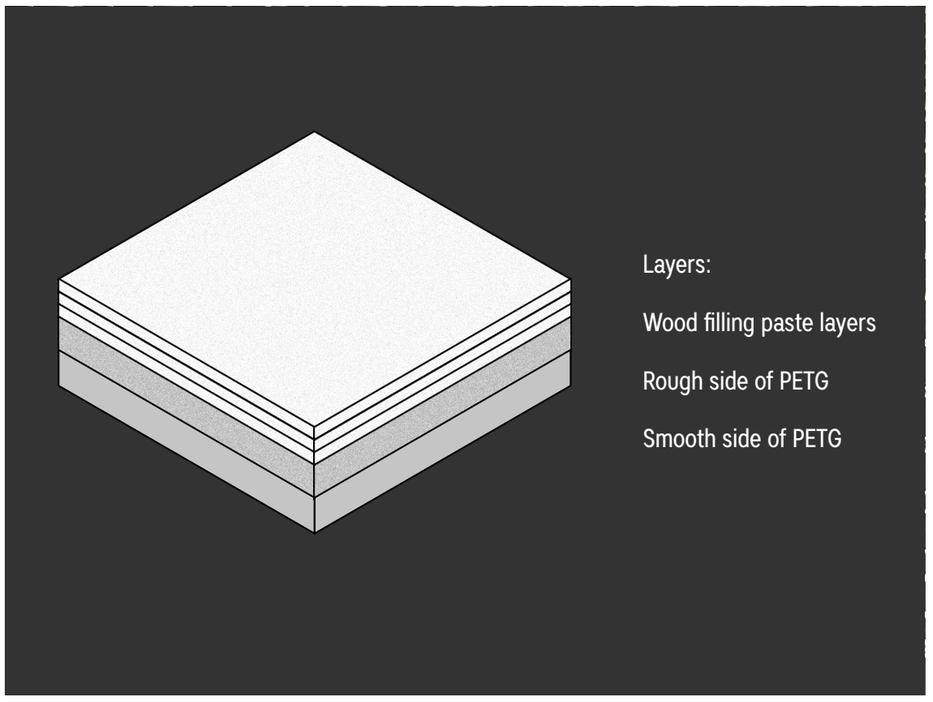
Wet

The solution is to dilute Wood Filling Paste and apply multiple layers.

Find a solution between ratios that is viscous enough to grip the texture, not too thin or watery, and doesn't require too many layers or dissolve when applying a new layer. At the same time, it should be able to be evenly applied on PETG panels with openings or thin stripes.

# Testing Material - Wall






3D Printed PETG on Rough Texture Printing Plate, coated with diluted Wood Filling Paste layers

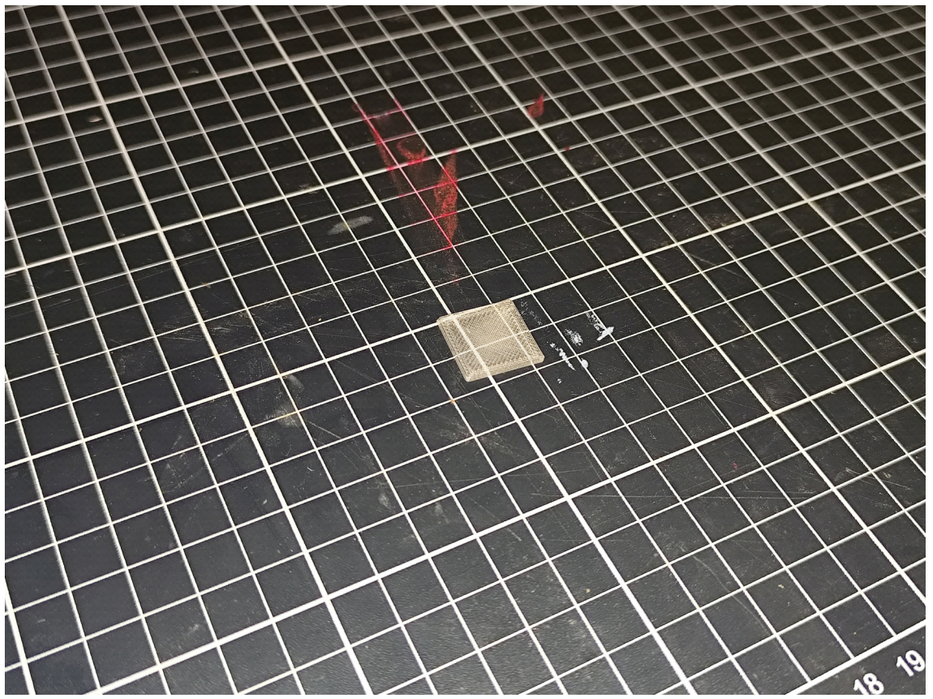
+ Inherits all of the benefits from the previous approach

+ Uniform thickness

+ Can apply in areas with holes and openings

- LONG drying time for each piece (days)

# Testing Material - Window



## 3D-print Transparent PETG Block

- Barely transparent

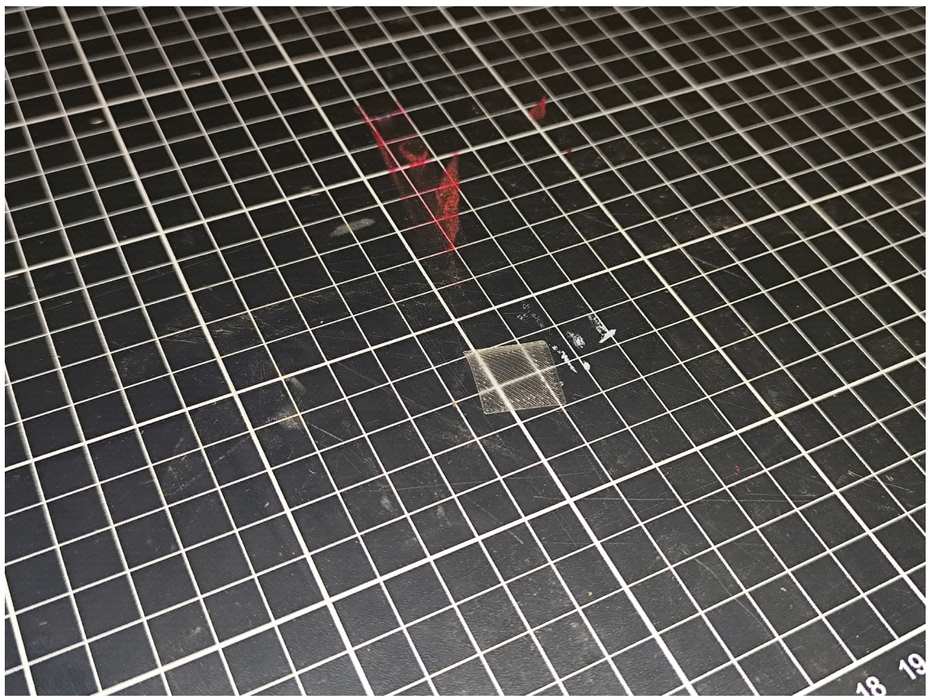
+ Hard

+ Easy to print; can print specific shapes

- Rough Surface, patterned texture

+ Bendable

# Testing Material - Window



3D-print Transparent PETG Sheet

- Somewhat transparent

+ Soft

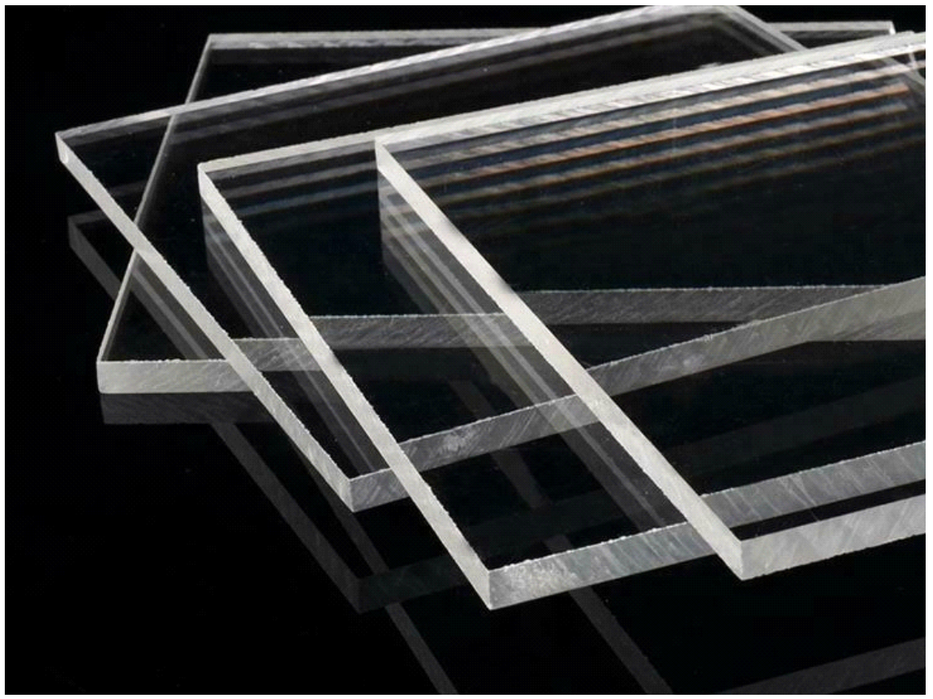
- Hard to work with, easily damaged when stripping off the printing plate

+ Easy to print; can print specific shapes

- Rough Surface, patterned texture

+ Bendable

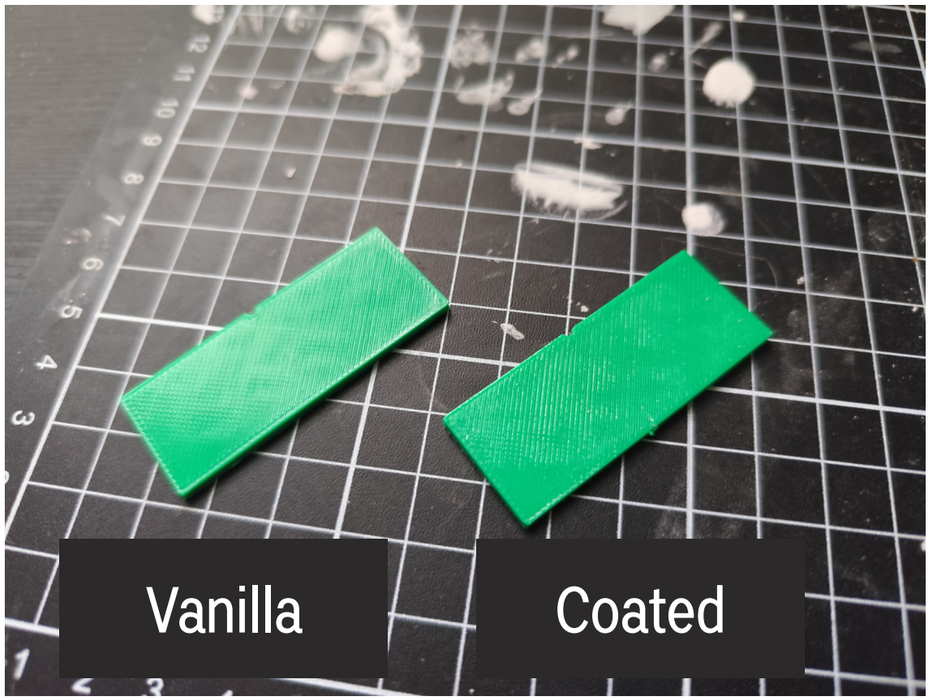
# Testing Material - Window

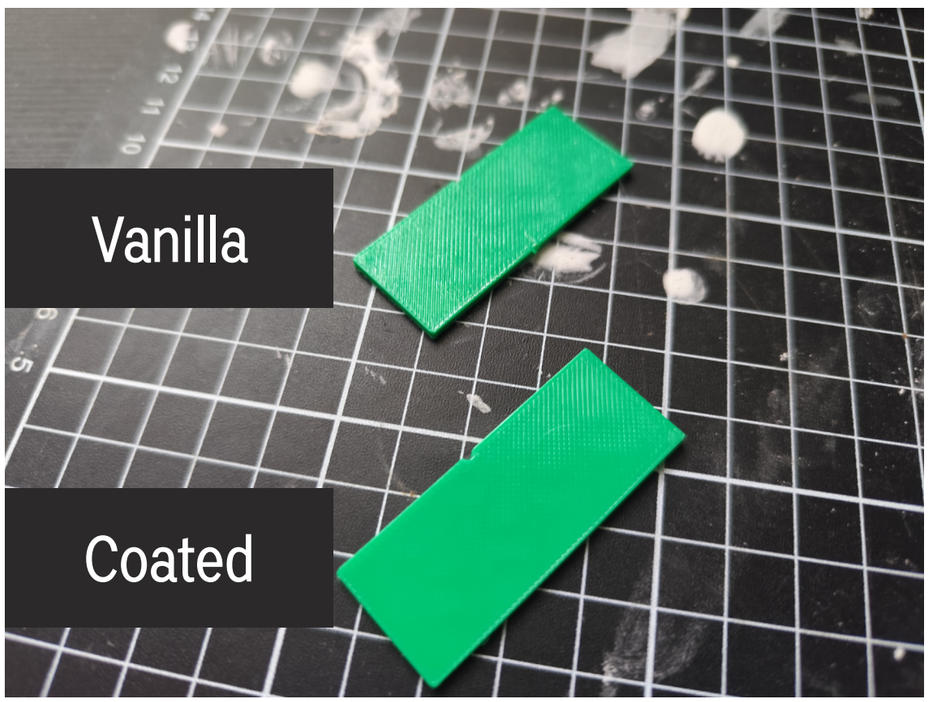


## Laser-Cut Acrylic Sheets

+ Very transparent
+ Hard
+ Can be precise with some tuning
+ Can cut out specific shapes
+ Smooth texture
+ Can be batch-produced through grid cutting.

# Testing Material - Framework




Varnish-Coated PETG

+ Inherits most of the benefits of PETG

- Some distance from the original colour

+ Varnish masks the 3D printer texture, slightly deeper colour

+ Minimum thickness coat; doesn't need to adjust part size.

+ Very adjustable

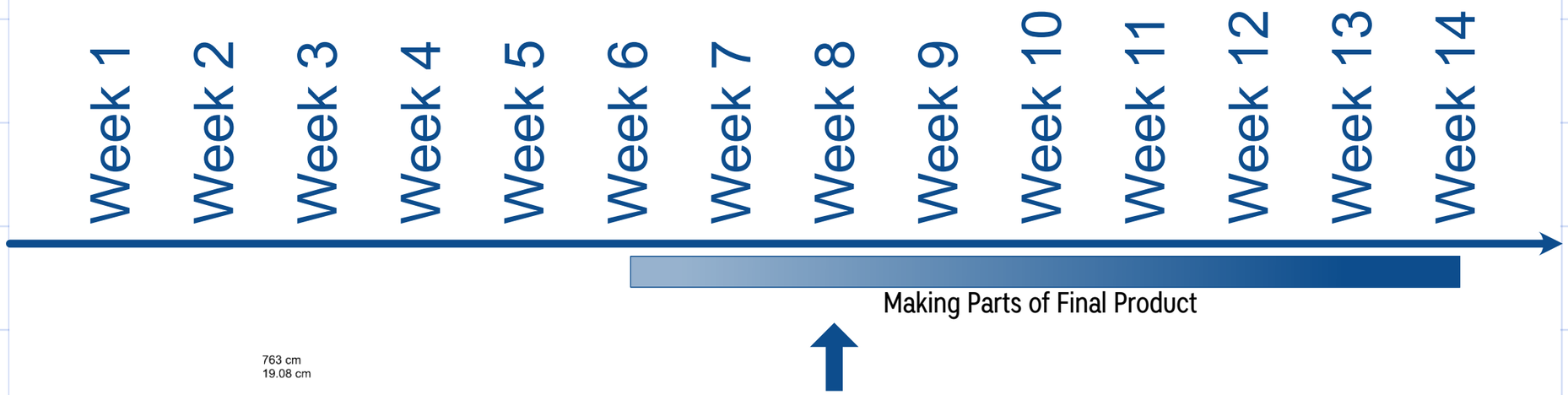
+ Can glue together easily with wood and acrylic

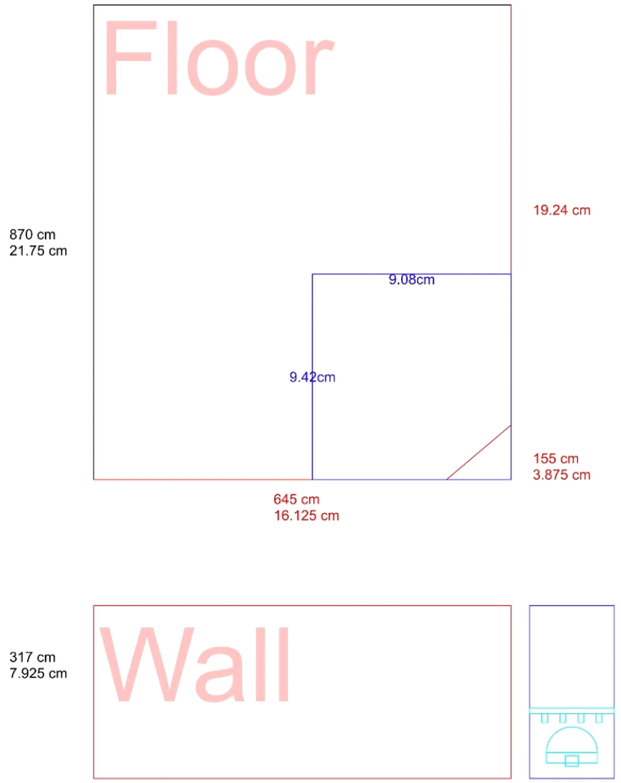
# Making Parts of Final Product

The general process of this section is as follows:

- Draw precise diameters of each piece in graphic software and export as vector files
- Model the piece in modelling software for 3D printing OR
- Convert to a laser cutting map in vector software
- Produce the piece
- Perform follow-up processing (painting, coating, layering-up, etc.)

# Graph of all pieces - Week 8

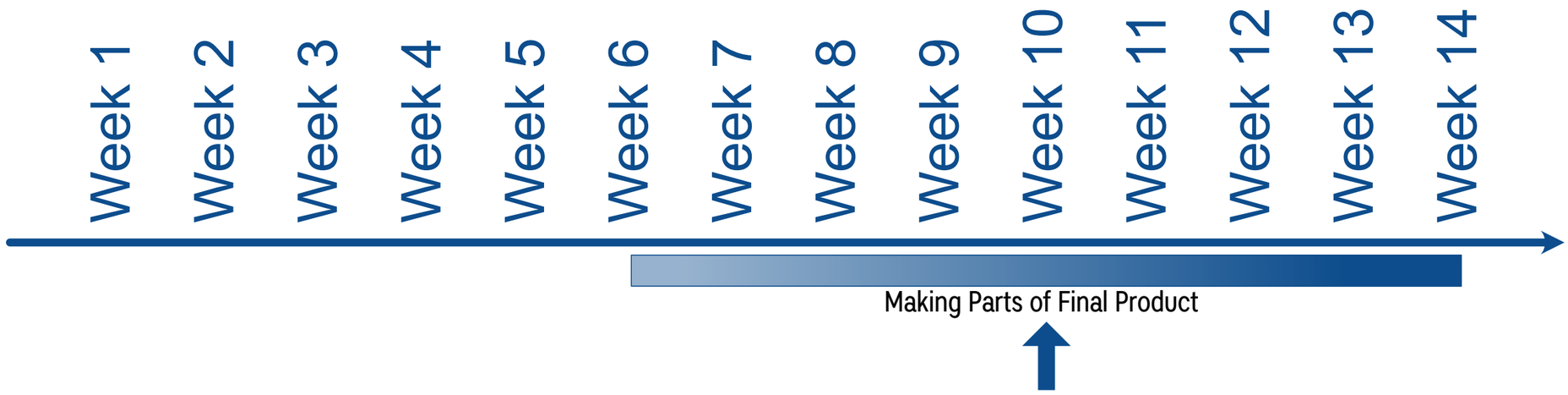


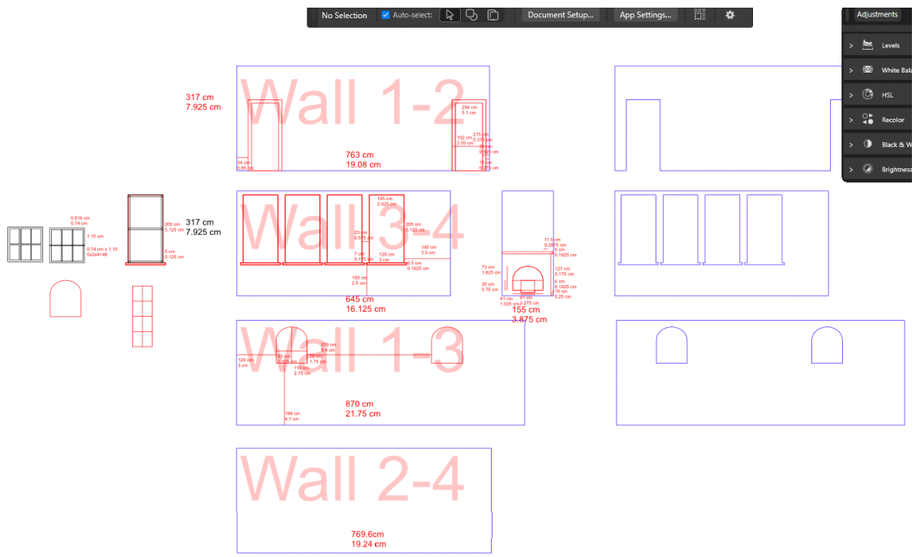

## Week 8 Graph

The making of the parts of the Final Product and the testing of the material were done in parallel while I was working. At this time, I had not yet tested the etched MDF Board, so there was an additional base layer at the bottom to support the floor layer. Therefore, the bottom layer was considered the most complex, and focused most at this time.

# Graph of all pieces - Week 10



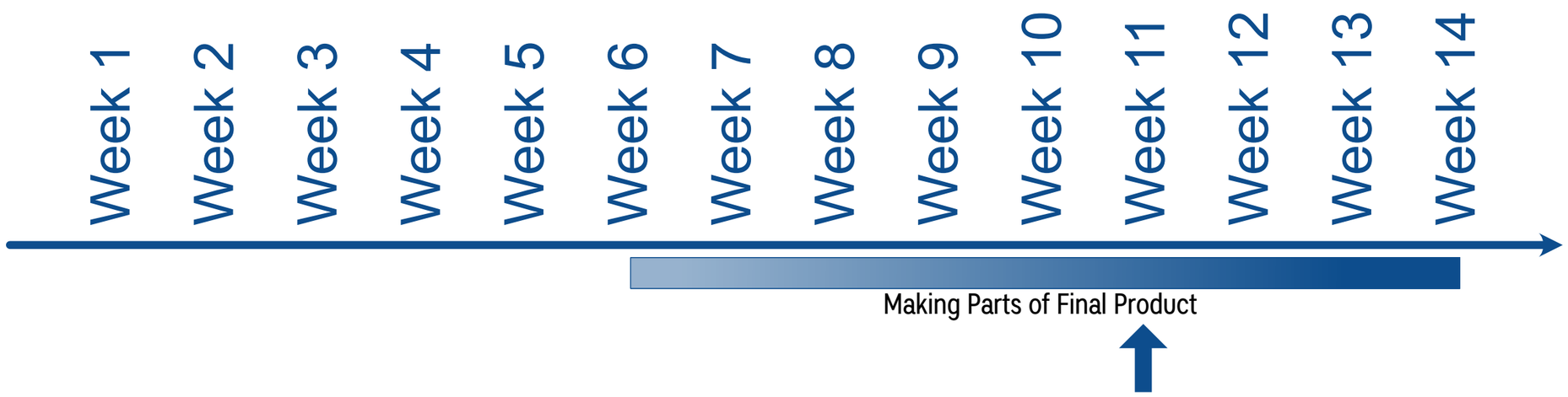

Week 10 Graph



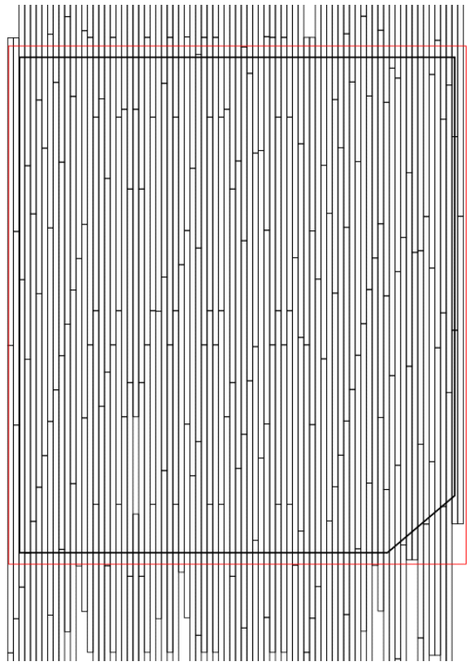

Temporarily ignore the floor plan due to the discovery of the Etched MDF Board; I no longer focus on the floor plate.

This is a wall plan made from a large number of precise measurements of classroom walls, including various data on windows, doors, and fireplaces.

# Graph of all pieces - Week 11



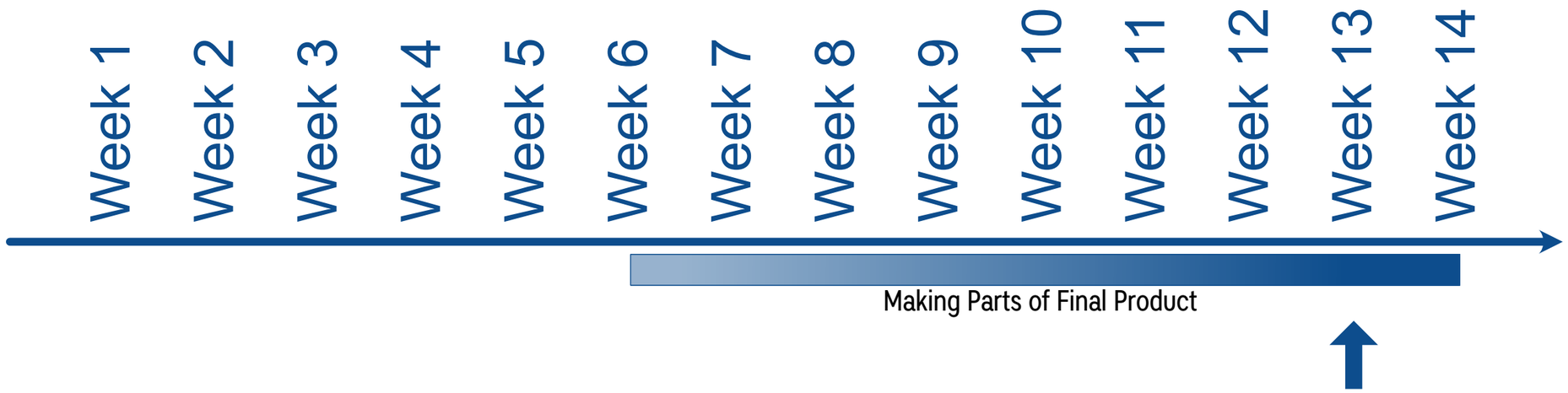

Week 11 Graph



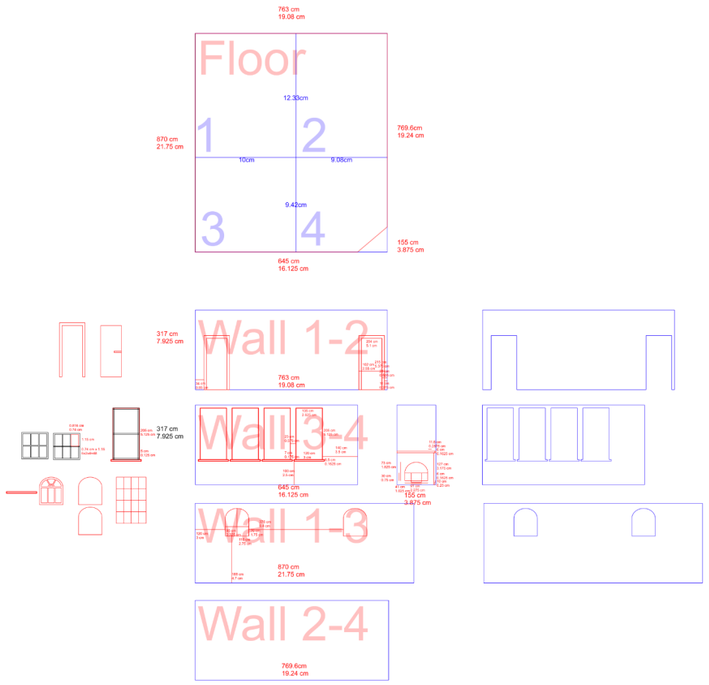
This is a map for the floor plate used for laser cutting, where the red lines are for cutting, and the black lines are for etching.

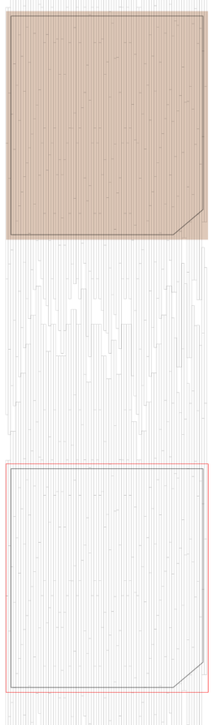
There are two weights of etching here: thinner lines are used to outline the border of the floor's wooden planks, and a thicker outline of etching is used to confirm the placement of the wall panels.

# Graph of all pieces - Week 13



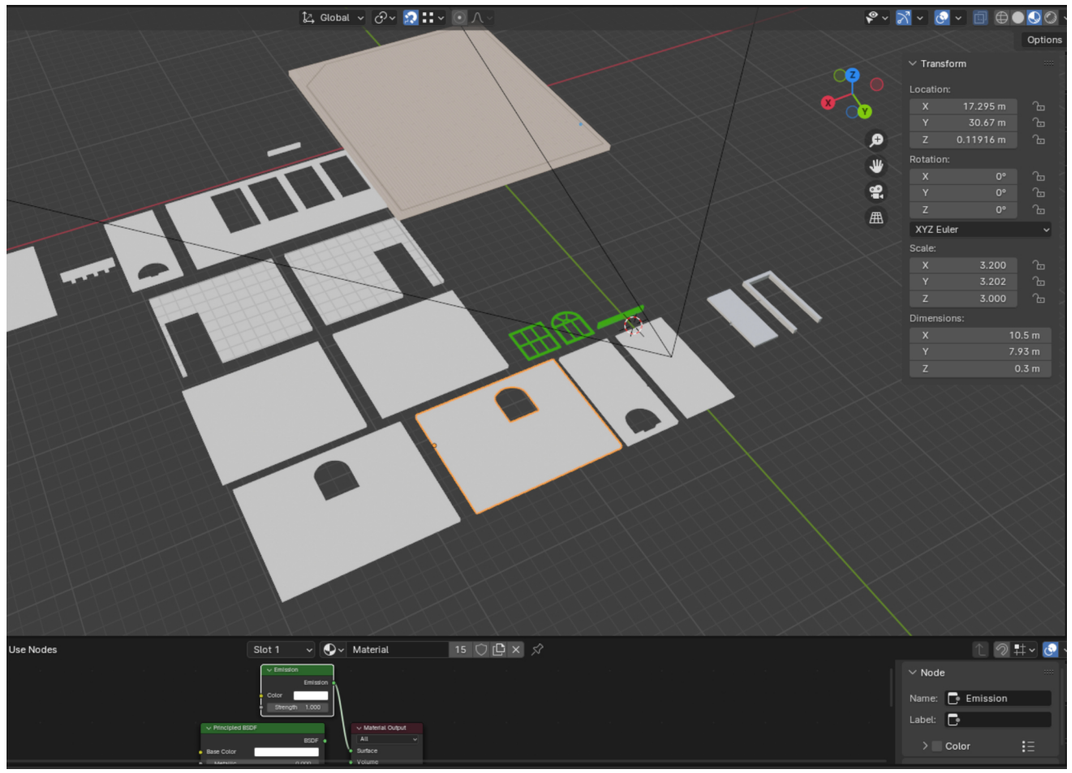

Week 13 Graph



Add/fix details on top of all previous graphs, such as door frames, specific window values, and the texture used to render the manual.

# Modeling Parts

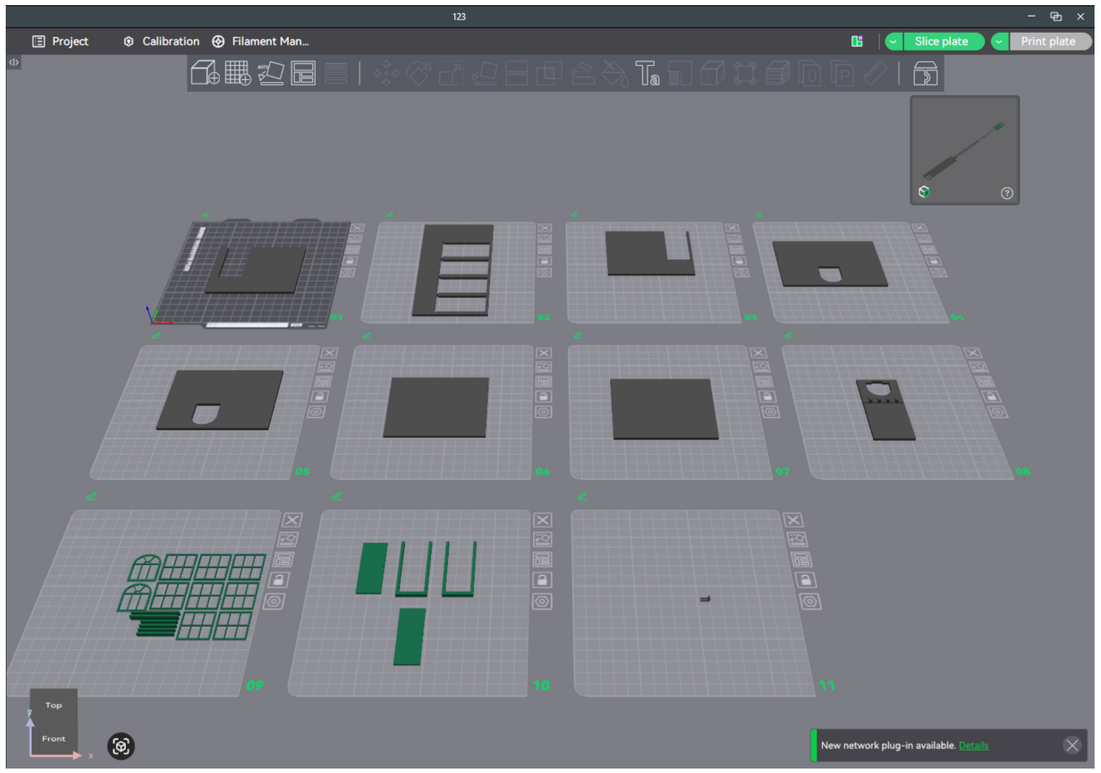



## Modeling Parts

Creating a model the size of the finished product to confirm that there are no issues with the assembly of parts and that everything fits properly.

The model is also exported for 3D printing, laser cutting, and the rendering used in the manual.

# 3D Printing



Failed 3D printing parts

## 3D Printing

Due to the large number of panel-type prints I have, these large and thin 3D-printed parts are very likely to warp or fail to hold their shape during printing, which has led to multiple modifications and iterations of the 3D printing configuration file.

At the same time, the initial model file was too large and thin, which caused some inevitable distortion when used for model assembly.

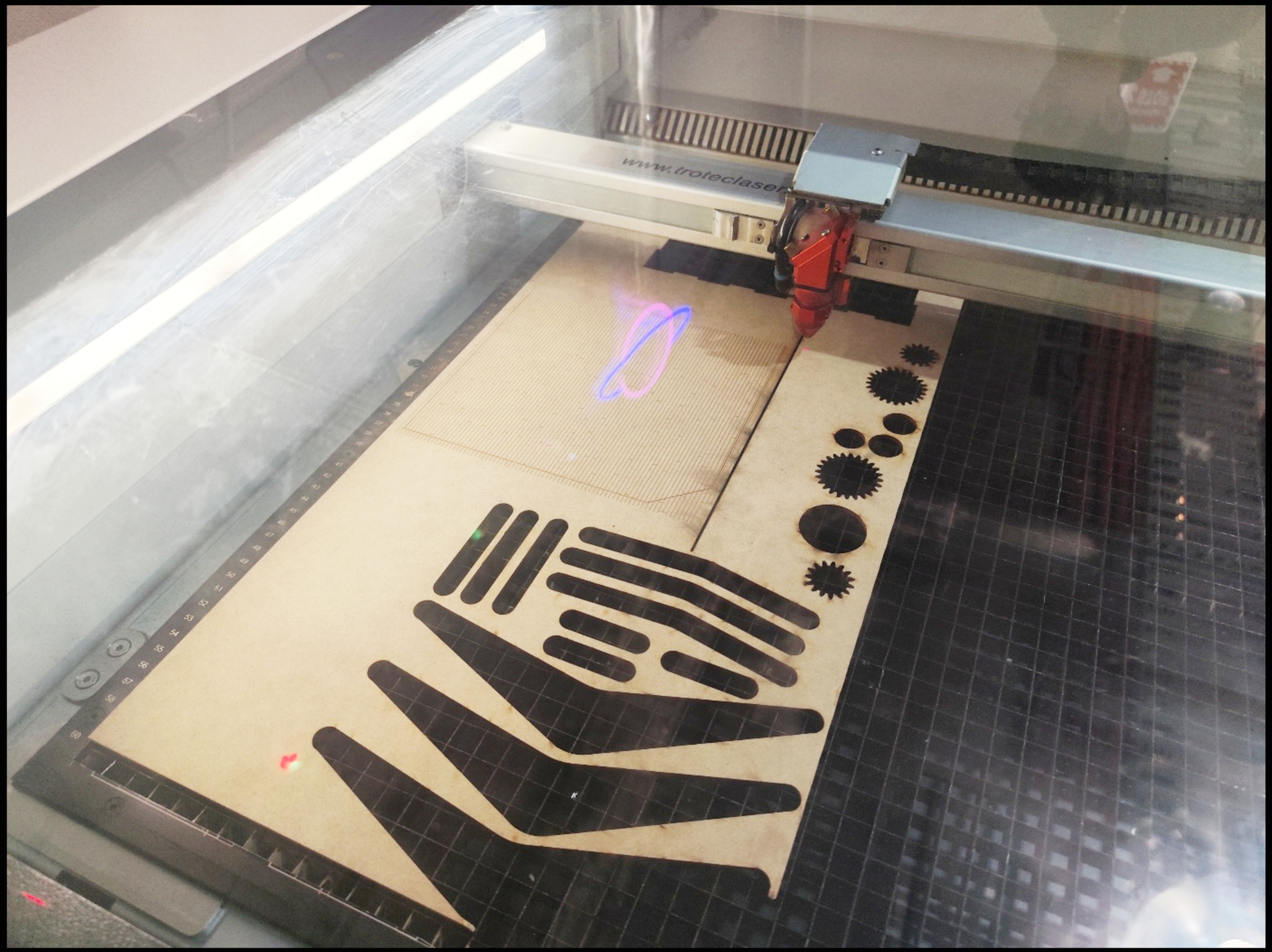
Later, thicker iterations were used for the final model.

The printing process is slowed down in order to reduce the rate of defective parts, resulting in a very long printing process.

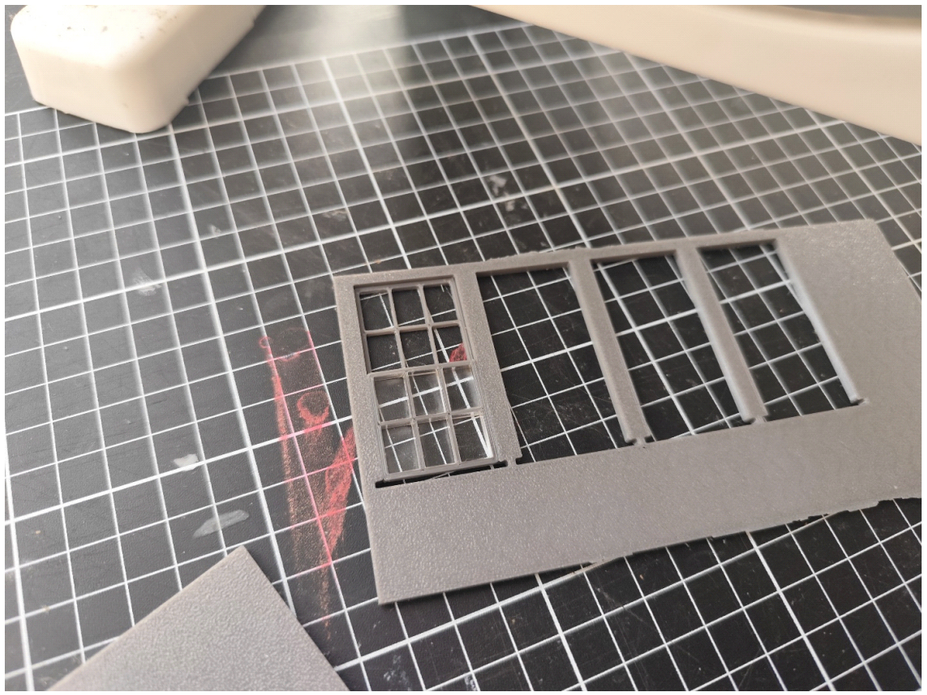
# Laser Cutting

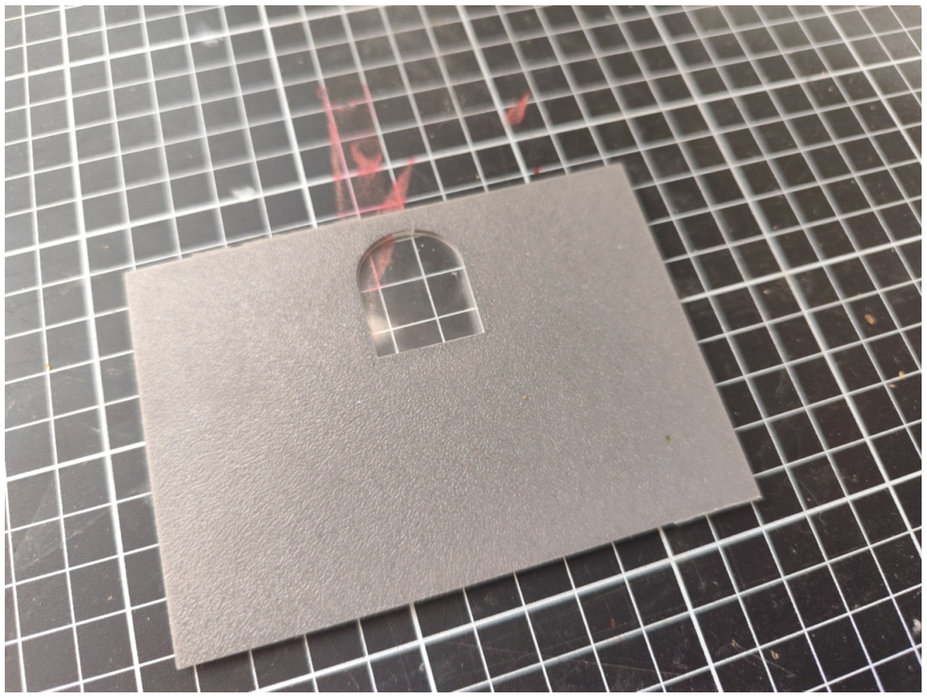
## Laser Cutting

Design a laser cutting map based on the opening size and laser width in the previous graph, and combine the cut pieces with the 3D-printed part.


www.troteclaser

# Combining

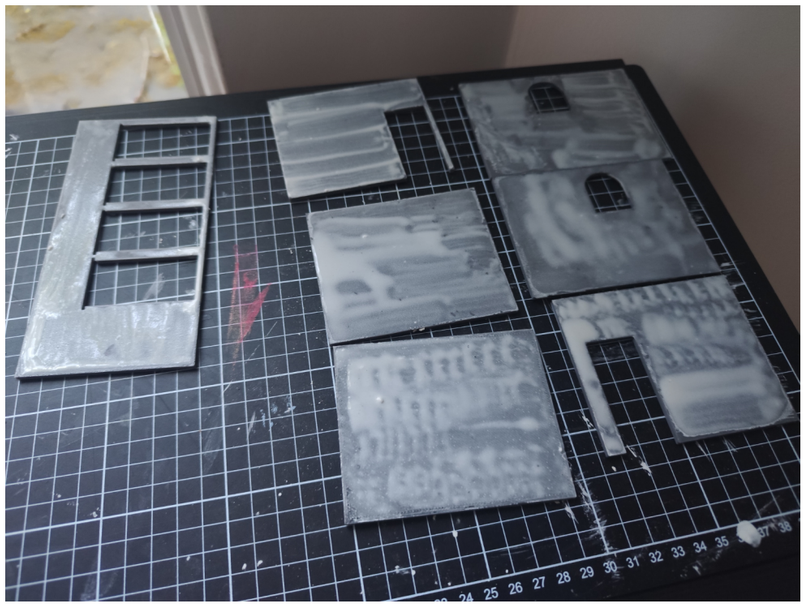




## Combining

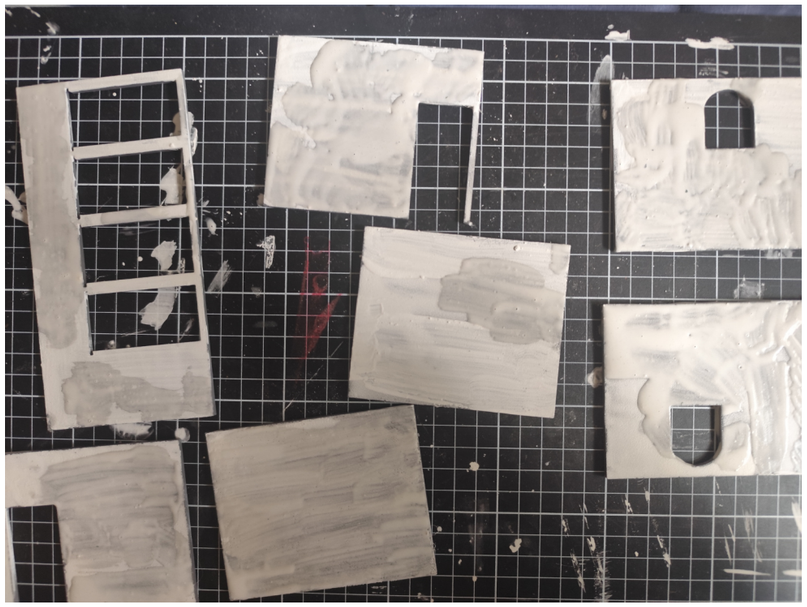
Testing combining cut pieces with the 3D-printed part to see if the combination is firm or not.

# Coating

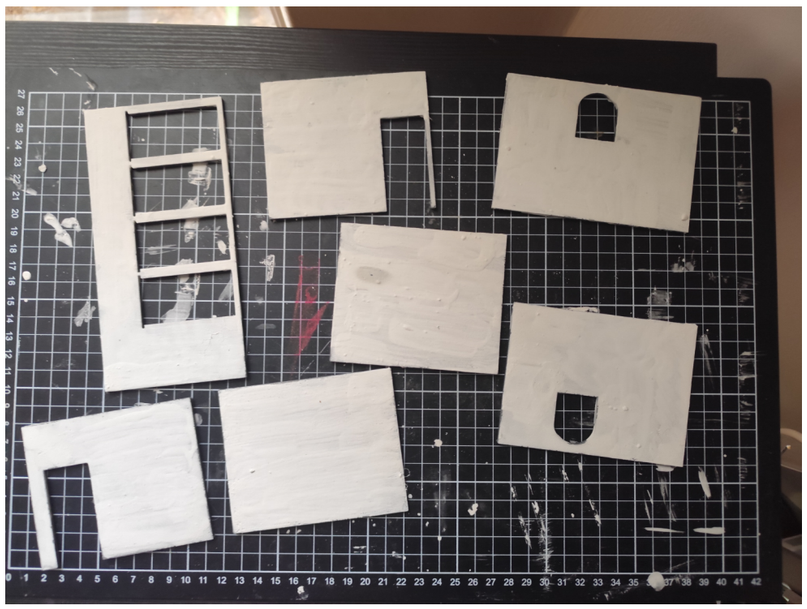


Coat 1

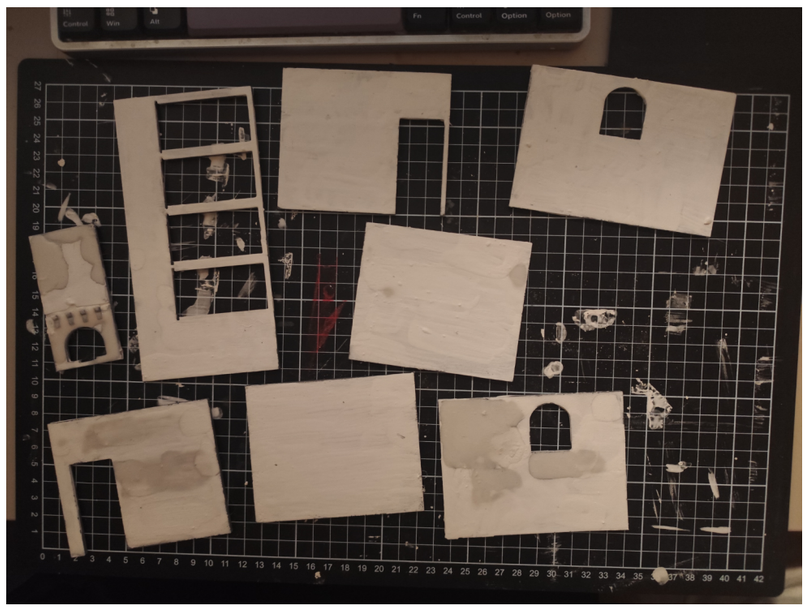
Coat 2



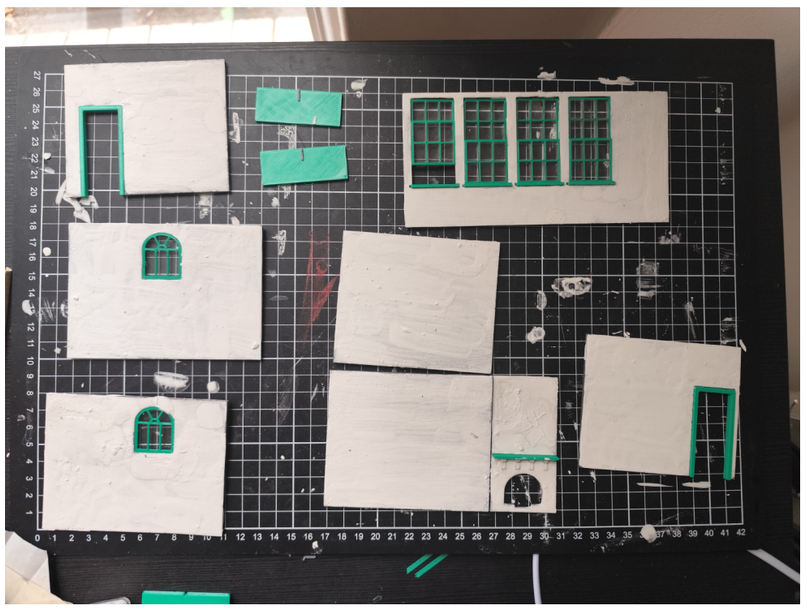
Coat 3



Coat 4

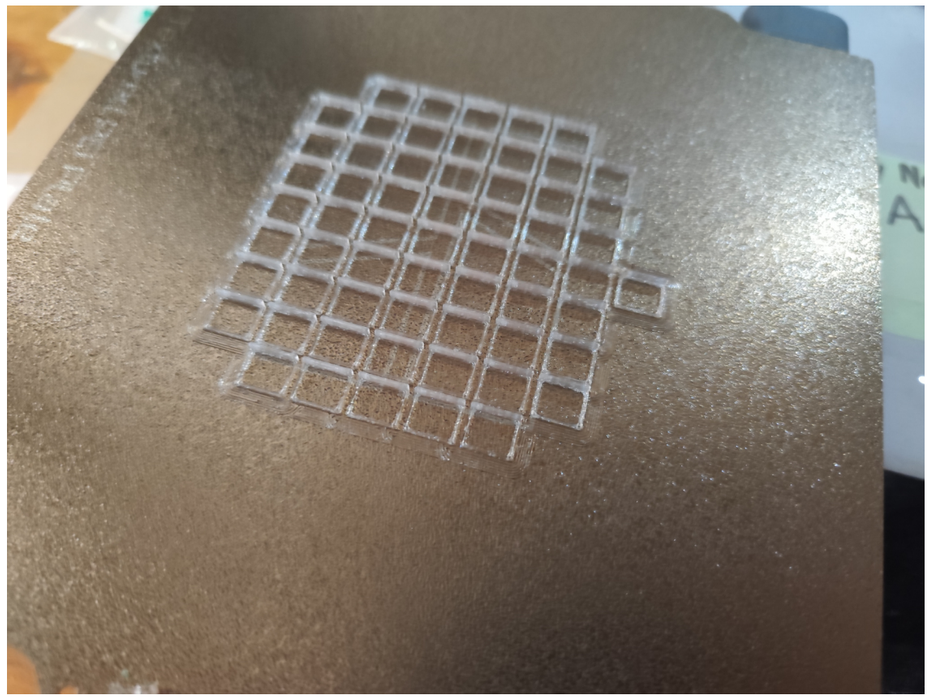


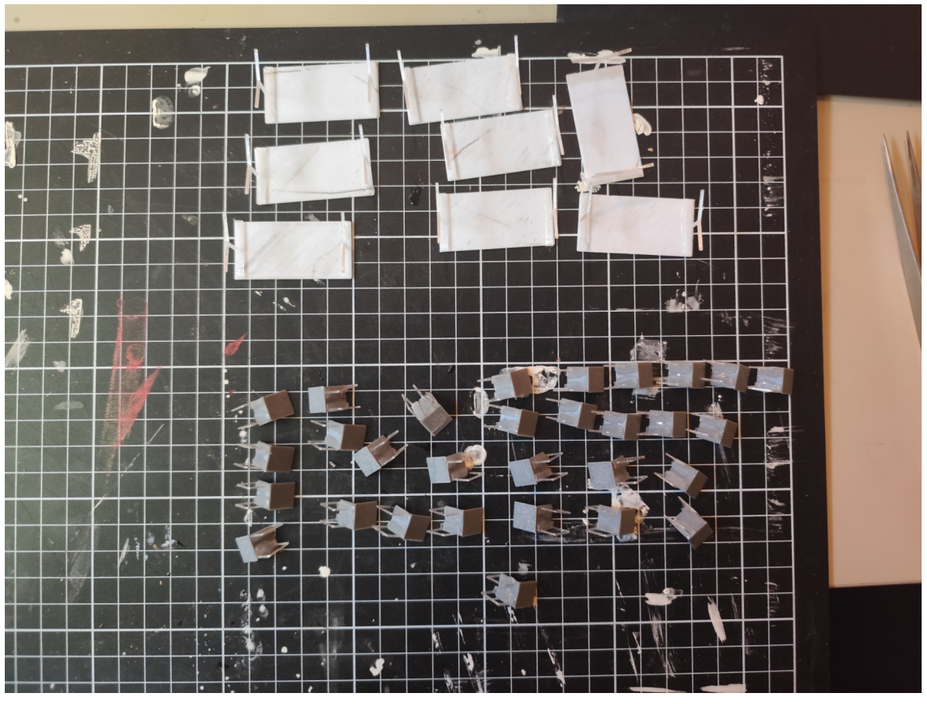
Coat 5



Coat 6 & Combining Parts

# Furnitures

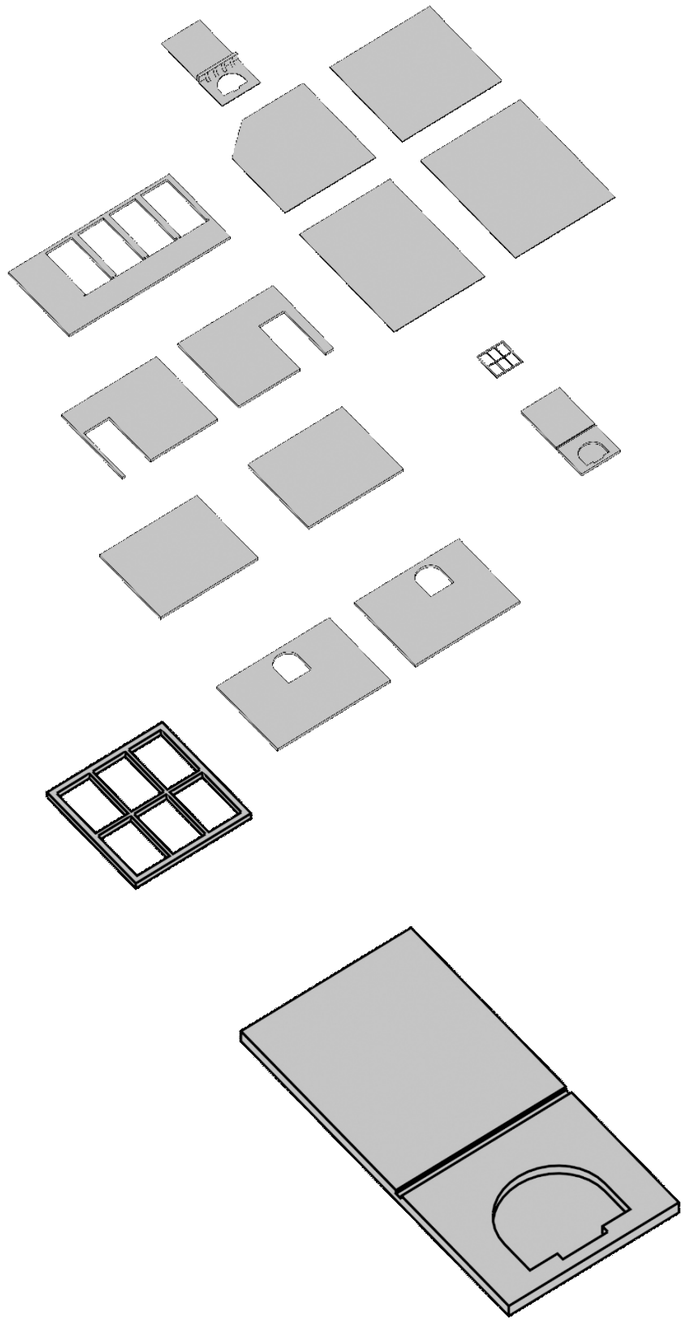




## Furnitures

In the final stage of the project, the classroom model alone still looked too plain, so furniture was added at the last moment, including tables and chairs, placed in the usual classroom positions.

# Manual



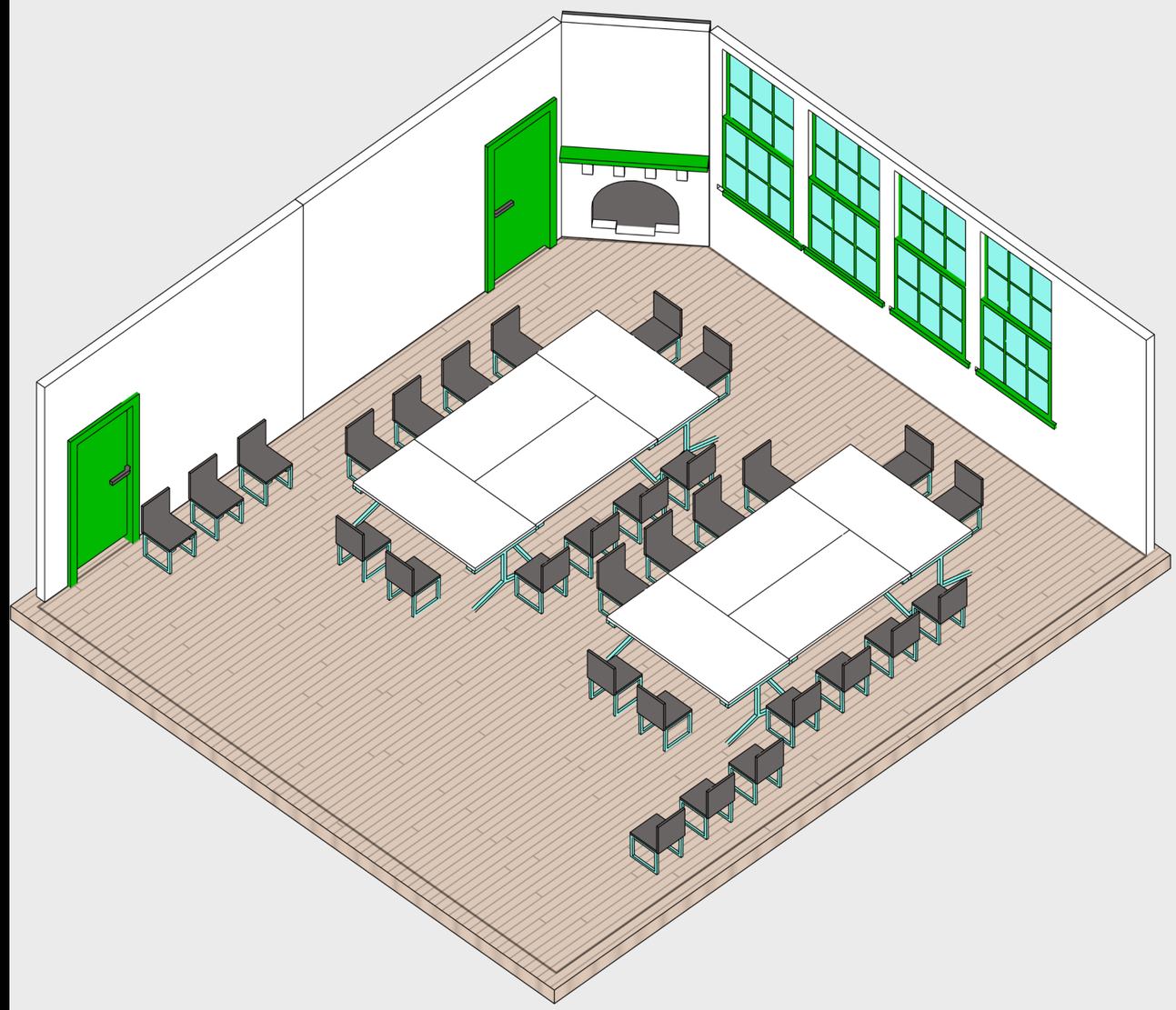
## Manual

I use Side Elevation and Exploded Assembly View similar to the LEGO manual.

These are rendered of me playing with angle outlines share a similar style to the exploded view in the LEGO manual.

# P7766 - The Classroom

## Manuel & Part List

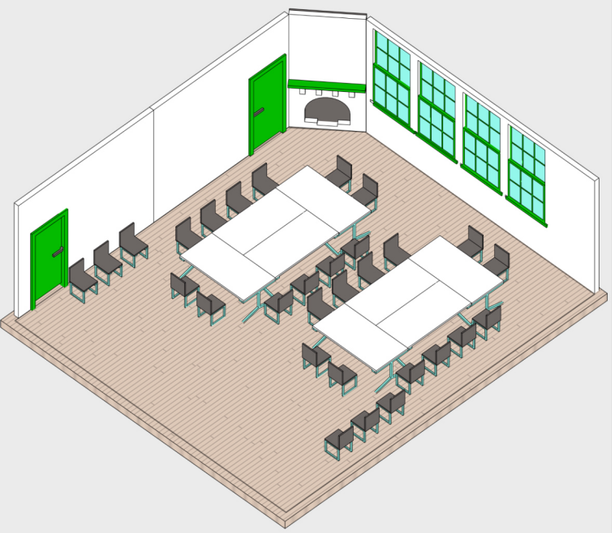


**Age 12 and above**
**170 seperate parts**

**Keep away from Children**
**Beware Children Swallowing**

# P7766 - The Classroom

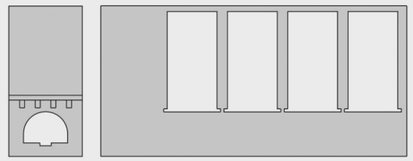
## Manuel & Part List



**Age 12 and above**
**170 seperate parts**

**Keep away from Children**
**Beware Children Swallowing**

In the box you should see:



P7766-WR-4-4



P7766-WR-3-4

x24

P7766-PT-WDO-3-4-T

x2

P7766-PT-WDO-1-3-T

x8

P7766-PT-WDO-3-4-F

x2

P7766-PT-WDO-1-3-F

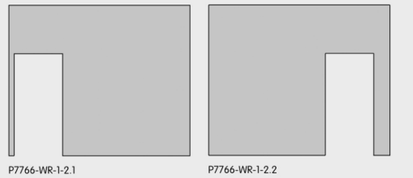


x4

P7766-PT-WDO-3-4-F-2



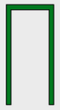
P7766-PT-WDO-3-4-F-2



P7766-WR-1-2.1



P7766-WR-1-2.2



x2

P7766-PT-D-F

x2

P7766-PT-D-D



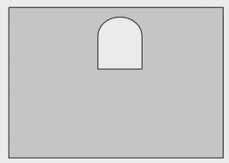
x2

P7766-PT-D-HDR

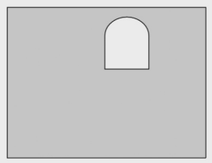
P7766-WR-2-4.1



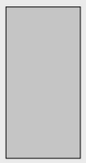
P7766-WR-2-4.2



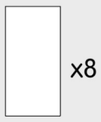
P7766-WR-1-3.1



P7766-WR-1-3.2



P7766-WR-4-4-BP



x8

P7766-DC-TBR-1

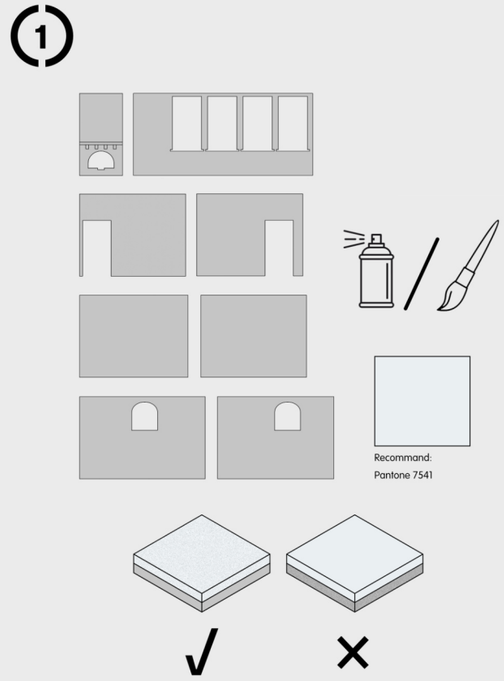
x16

P7766-DC-TBR-2

x60 x30

P7766-DC-CHR-1 P7766-DC-CHR-2

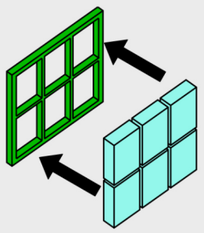
P7766-BS-01



Recommand:
Pantone 7541

Apply paint to the rough surface of the wall panels.

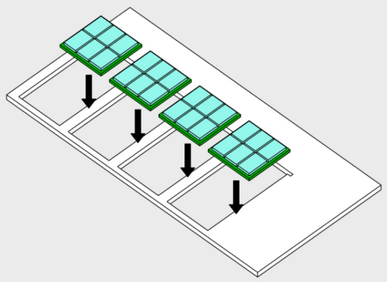
2



x8

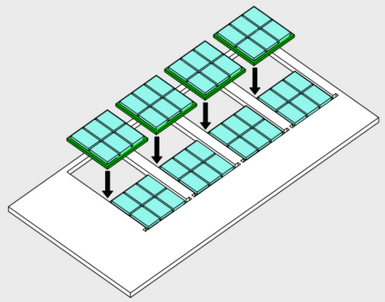
Insert the transparent block P7766-PT-WDO-3-4-T into the directional window frame P7766-PT-WDO-3-4-F

3



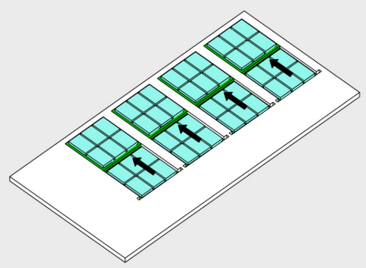
Insert the window frame into the bottom of the opening of P7766-WR-3-4
This is the back side of P7766-WR-3-4 panel

4



Insert the window frame into the top of the opening of P7766-WR-3-4
This is the back side of P7766-WR-3-4 panel

5

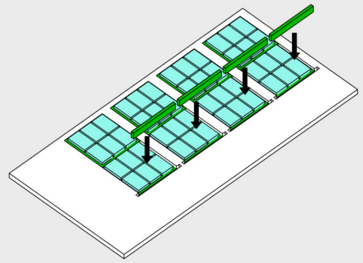


Before

After

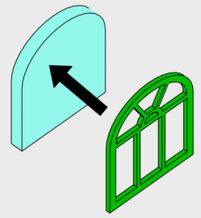
Push inserted window frames up

6



Insert the windowsill P7766-PT-WDO-3-4-F-2 into the bottom opening of the P7766-WR-3-4.

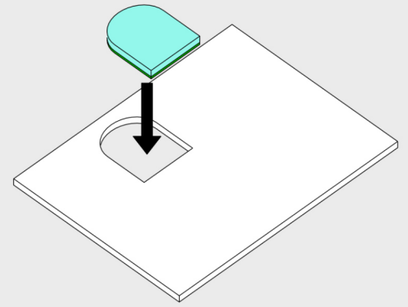
7



x2

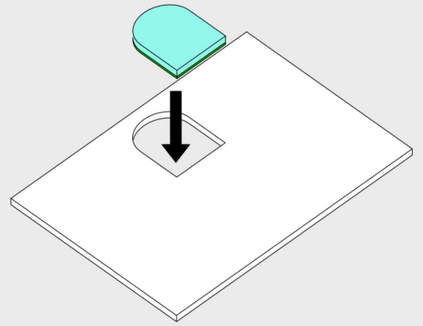
Layup transparent curve window block P7766-PT-WDO-1-3-T with
curve window frame P7766-PT-WDO-1-3-F

8



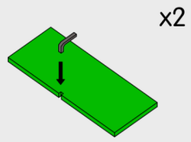
Insert the curve window frame into the opening of P7766-WR-1-3.2
This is the back side of P7766-WR-1-3.2 panel

9



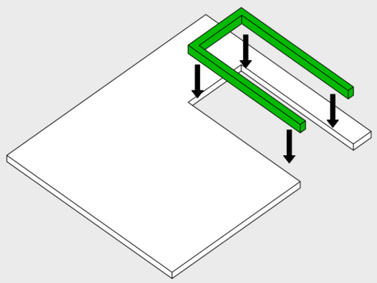
Insert the curve window frame into the opening of P7766-WR-1-3.1
This is the back side of P7766-WR-1-3.1 panel

10



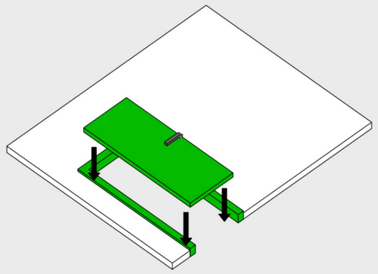

Insert the door handle P7766-PT-D-HDR to door part P7766-PT-D-D

11



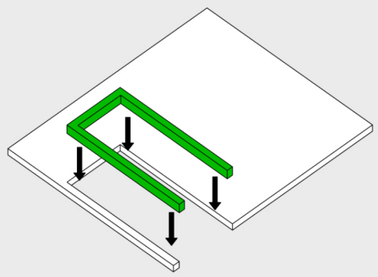
Insert door frame part P7766-PT-D-F to P7766-WR-1-2.2
This is the back side of P7766-WR-1-2.2 panel

12



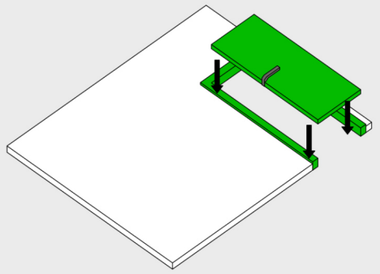
Insert door with handle to P7766-WR-1-2.2 panel

13



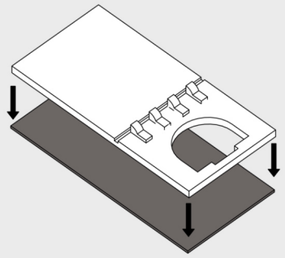
Insert door frame part P7766-PT-D-F to P7766-WR-1-2.1
This is the back side of P7766-WR-1-2.1 panel

14



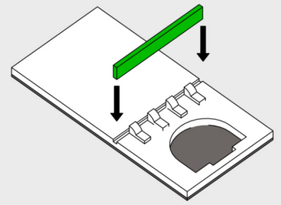
Insert door with handle to P7766-WR-1-2.2 panel

15



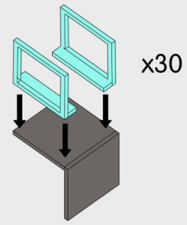
Layup panel P7766-WR-4-4 with back panel P7766-WR-4-4-BP

16



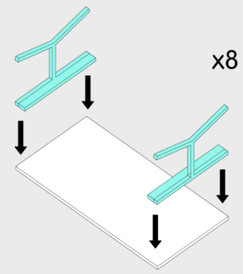
Insert mantelpiece P7766-PT-WDO-3-4-F-2 to panel

17



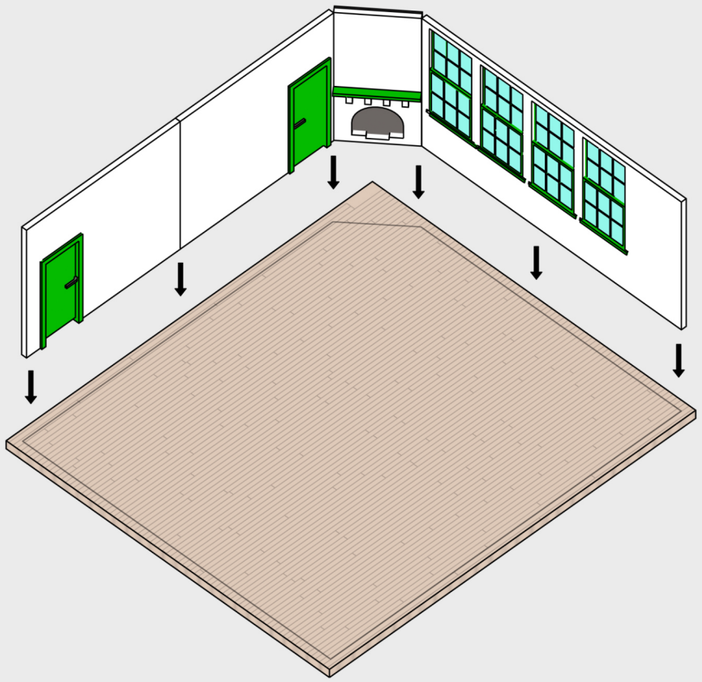

Combining chair body P7766-DC-CHR-1 and chair leg P7766-DC-CHR-1

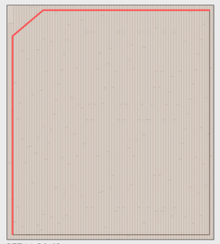
18




Combining table board P7766-DC-TBR-1 and table leg P7766-DC-TBR-2

19

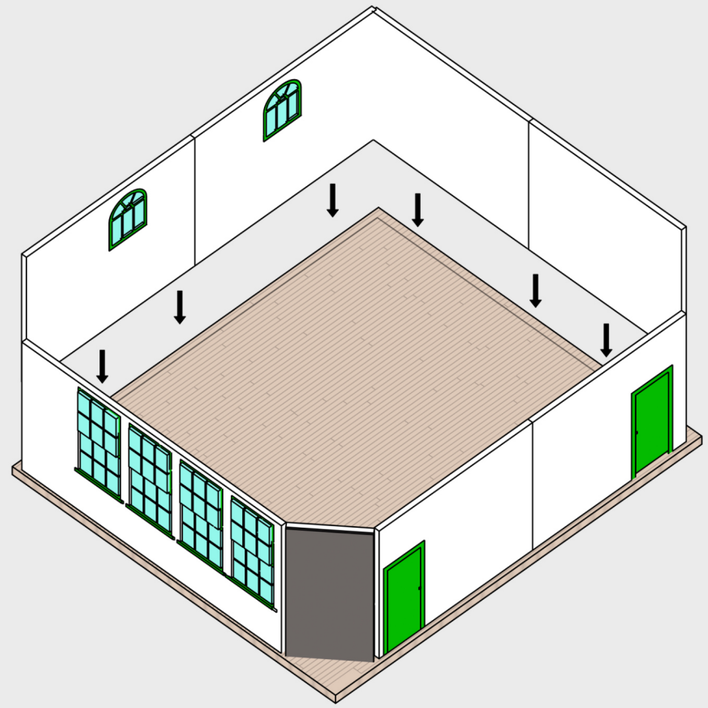


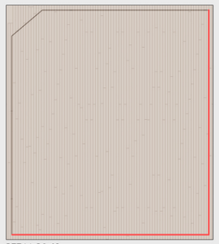


P7766-BS-01

Sticking wall panels on the floor plane

20

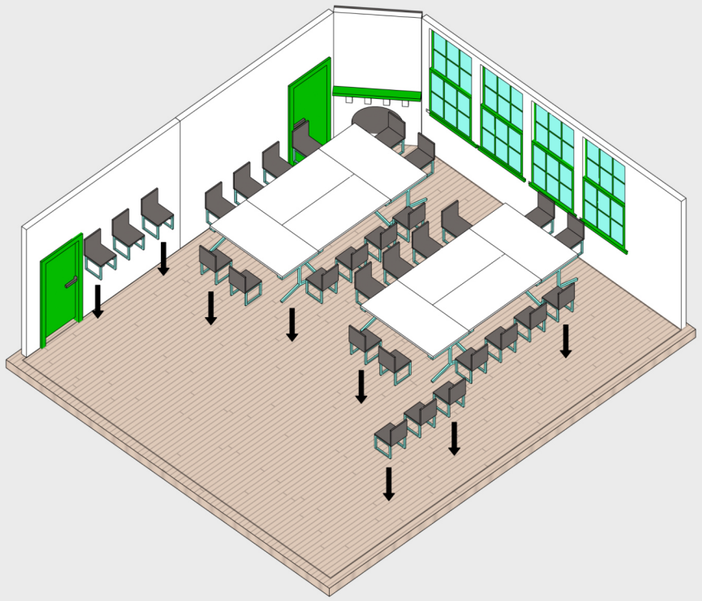




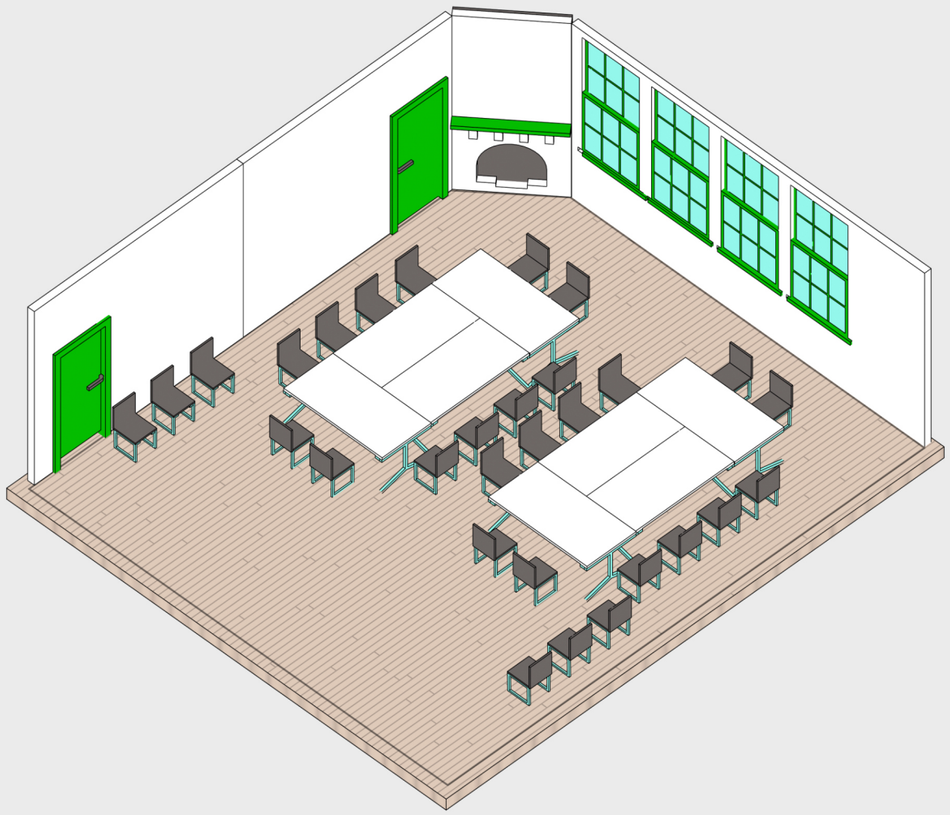
P7766-BS-01

Sticking wall panels on the floor plane

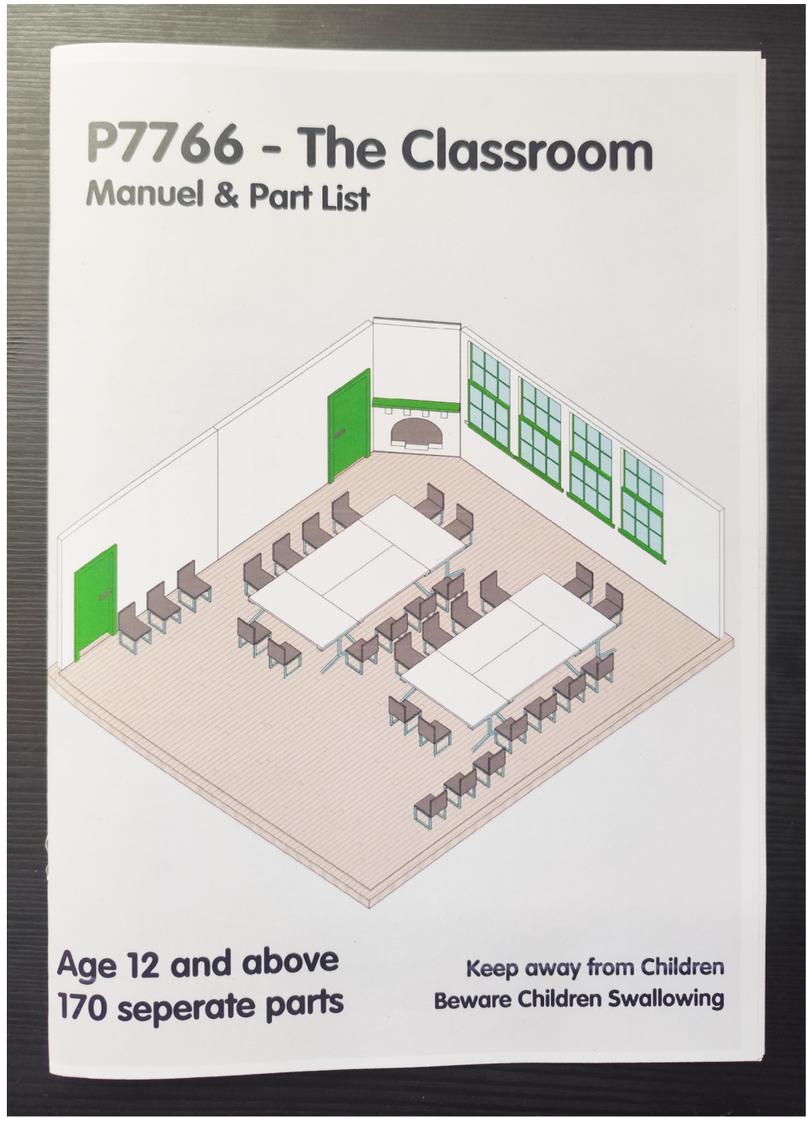
21



Adding tables and chairs

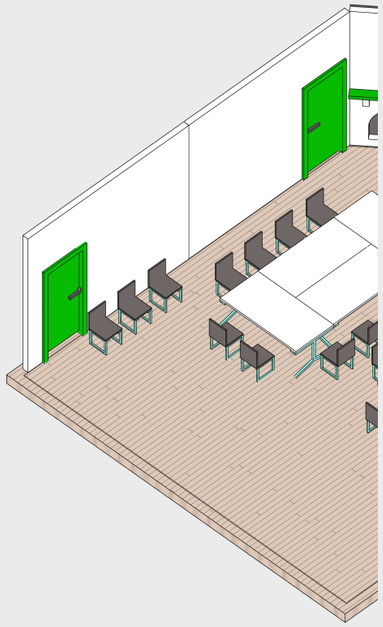


# Manual - Printing

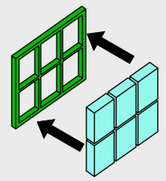



## Manual - Printing

When binding a manual into a booklet physically, it is necessary to rearrange A4 pages into an A3-size printing format using Book arrangement.



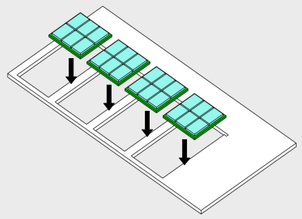
2



x8

Insert the transparent block P7766-PT-WDO-3-4-T into the directional window frame P7766-PT-WDO-3-4-F

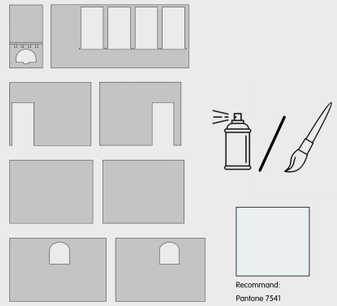
3



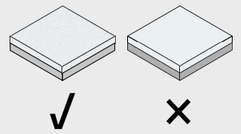
Insert the window frame into the bottom of the opening of P7766-WR-3-4
This is the back side of P7766-WR-3-4 panel

&

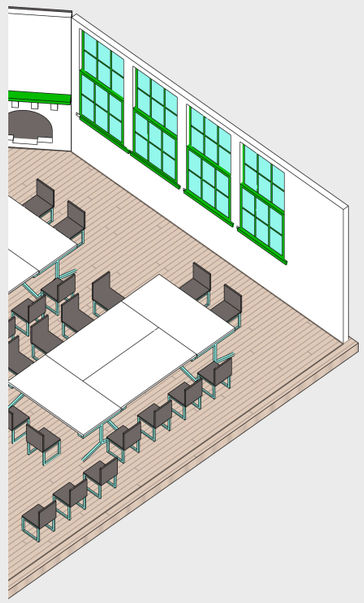
1

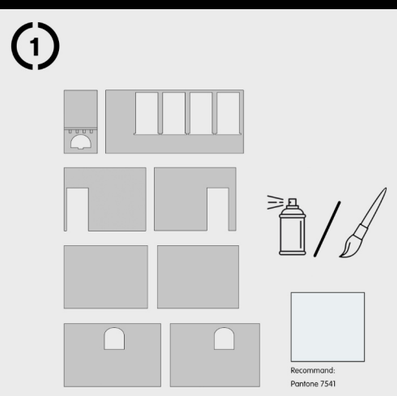


Recommand:
Pantone 7541

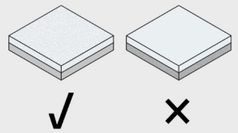


Apply paint to the rough surface of the wall panels.



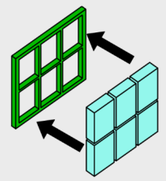


Recommand:
Pantone 7541



Apply paint to the rough surface of the wall panels.

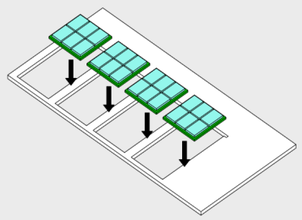
2



x8

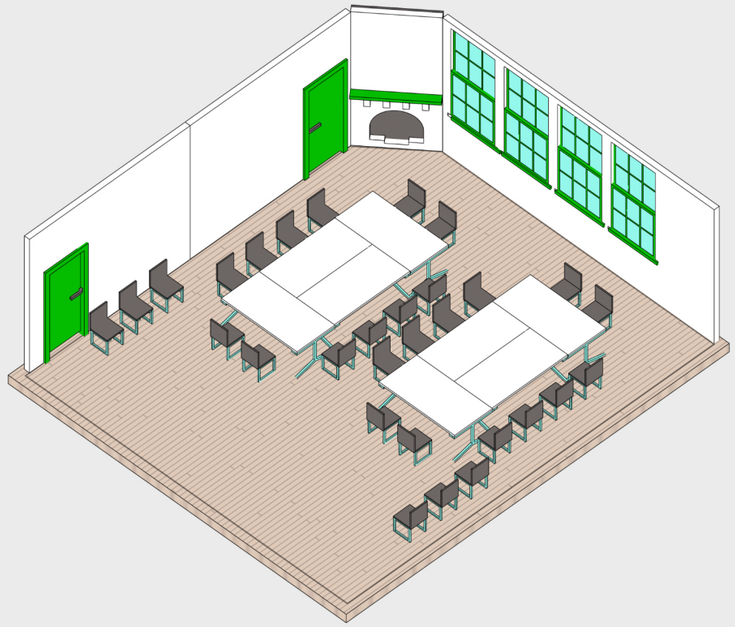
Insert the transparent block P7766-PT-WDO-3-4-T into the directional window frame P7766-PT-WDO-3-4-F

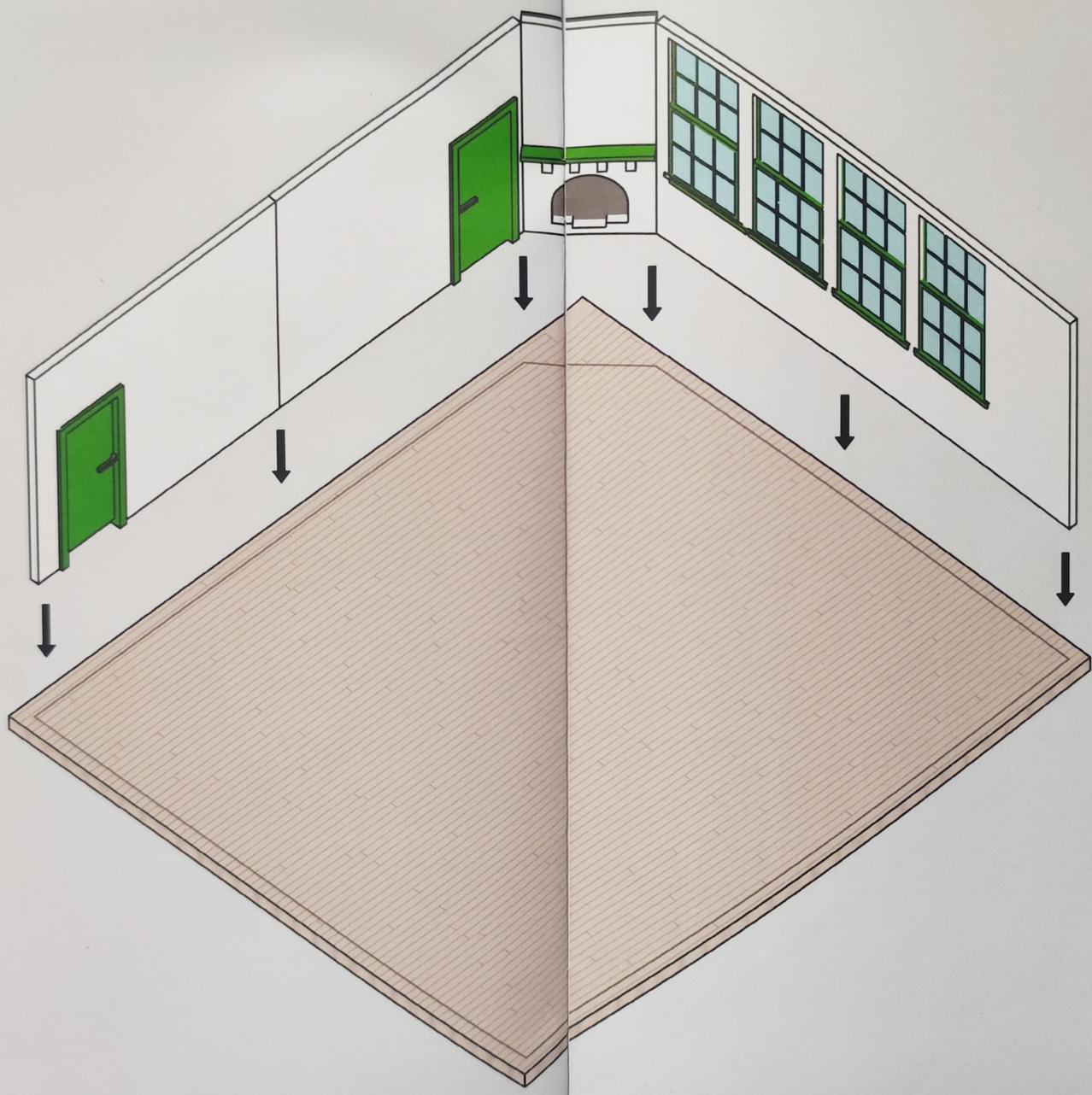
3



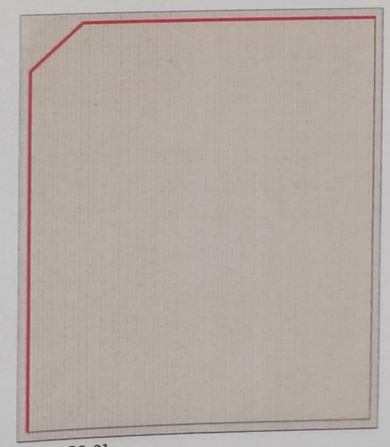
Insert the window frame into the bottom of the opening of P7766-WR-3-4
This is the back side of P7766-WR-3-4 panel

&



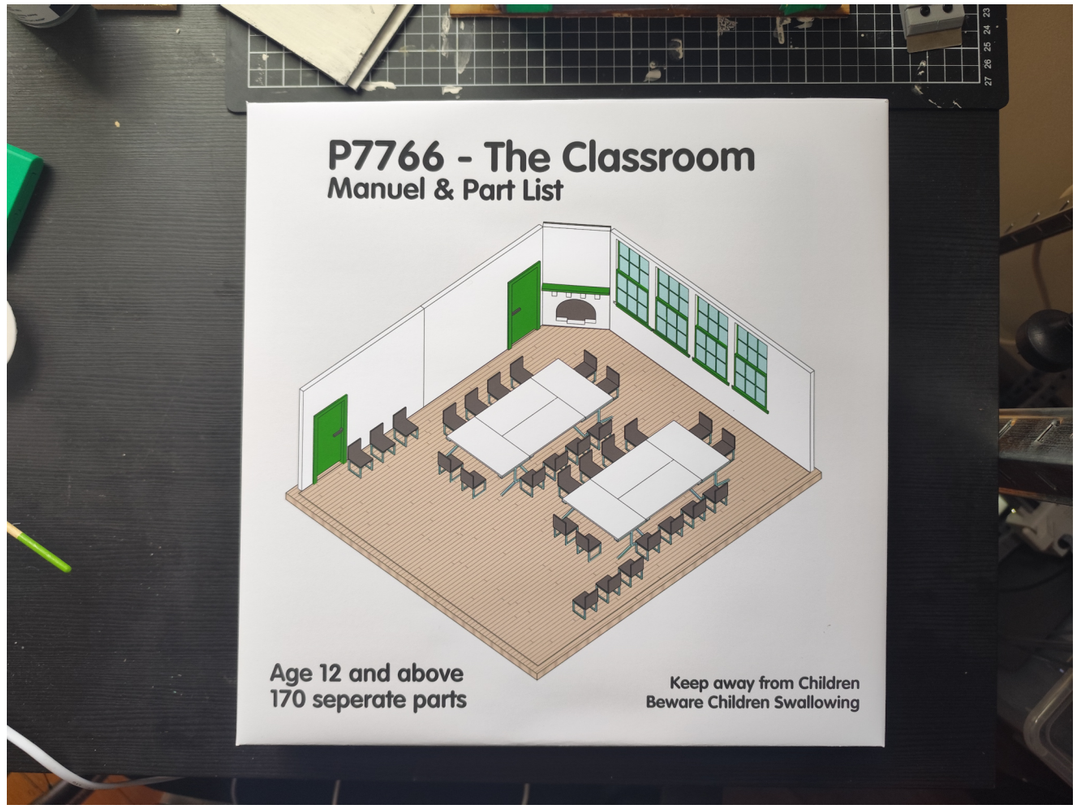


Sticking wall panels on the floor plane



P7766-BS-01

# Box



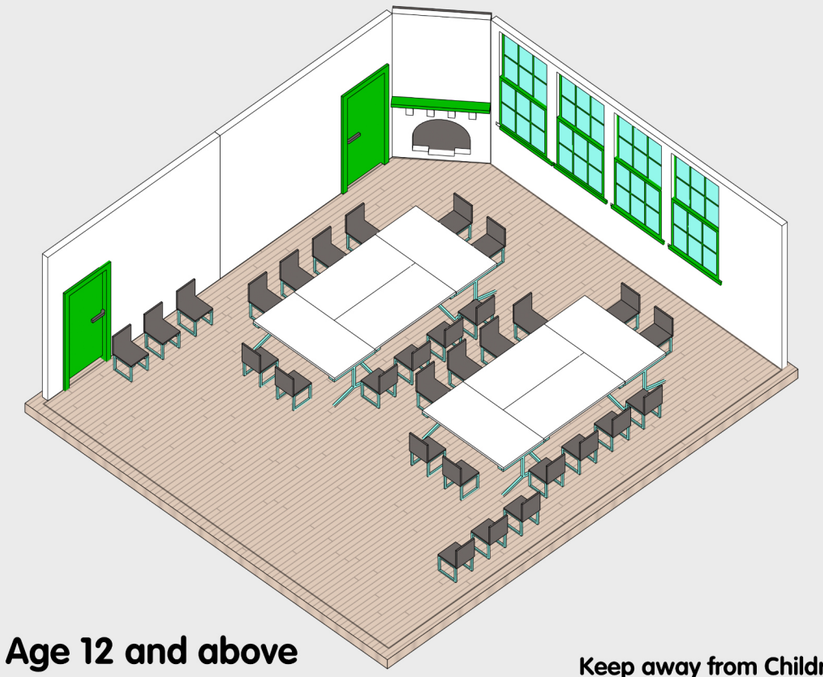

## Box

I plan to use Side Elevation and Exploded Assembly View similar to the LEGO manual.

These rendered pictures of me playing with angle outlines share a similar style to the exploded view in the LEGO manual.

# P7766 - The Classroom

## Manuel & Part List

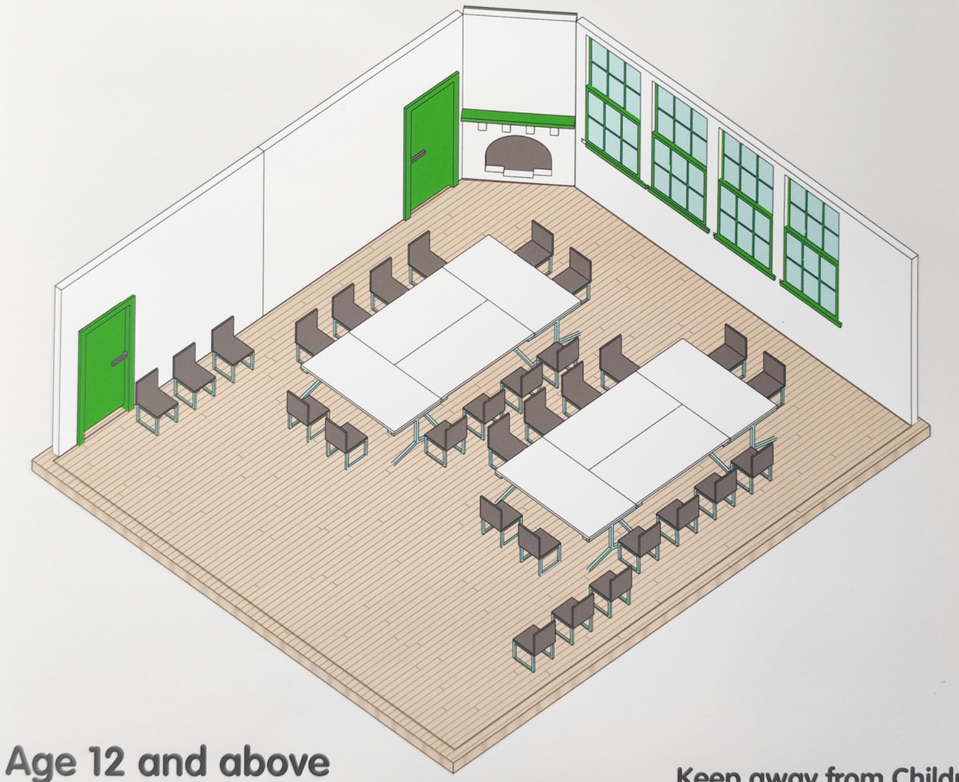


**Age 12 and above**
**170 seperate parts**

**Keep away from Children**
**Beware Children Swallowing**

# P7766 - The Classroom
## Manuel & Part List

# P7766 - The Classroom
## Manuel & Part List



**Age 12 and above**
**170 seperate parts**

**Keep away from Children**
**Beware Children Swallowing**

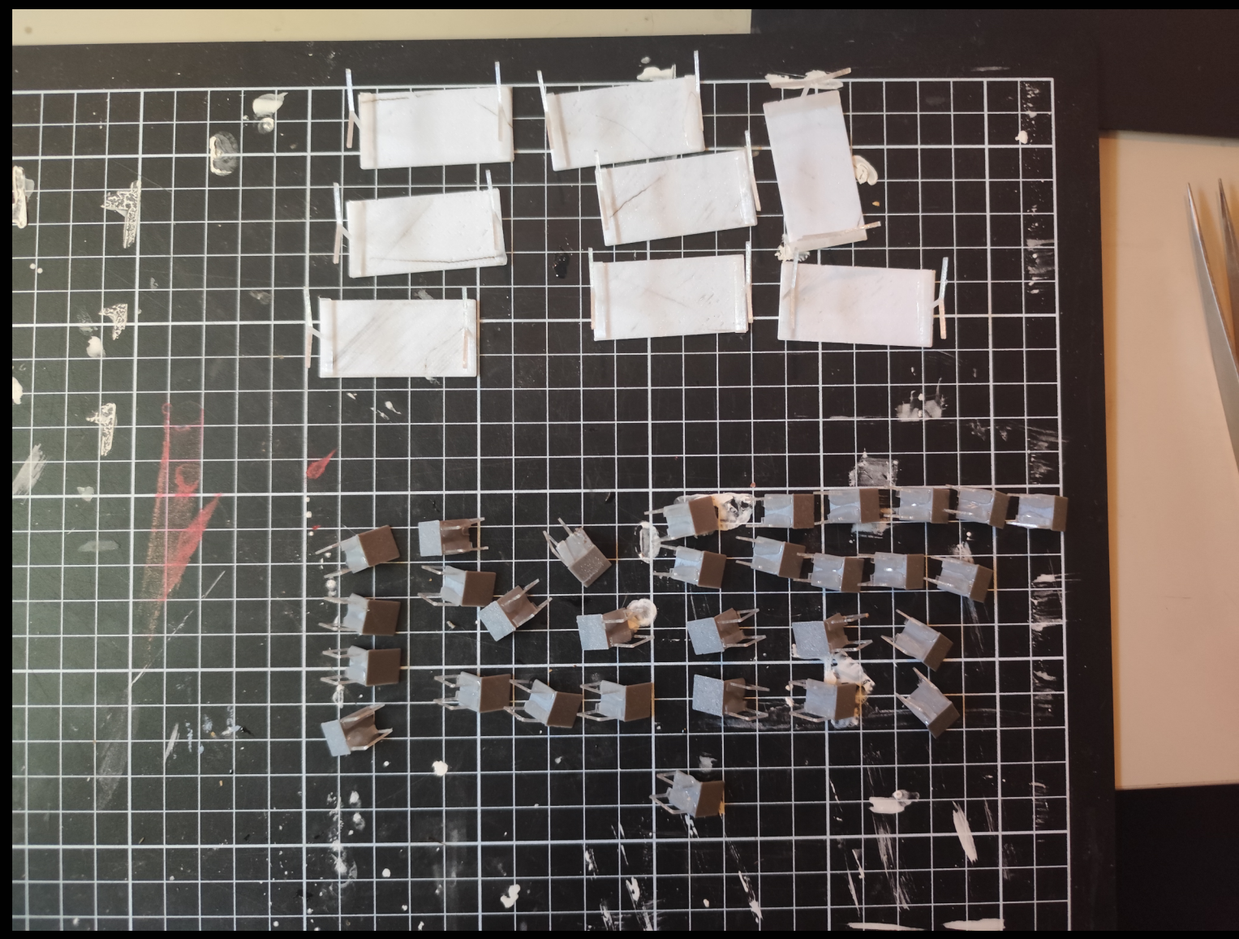
# Assemble

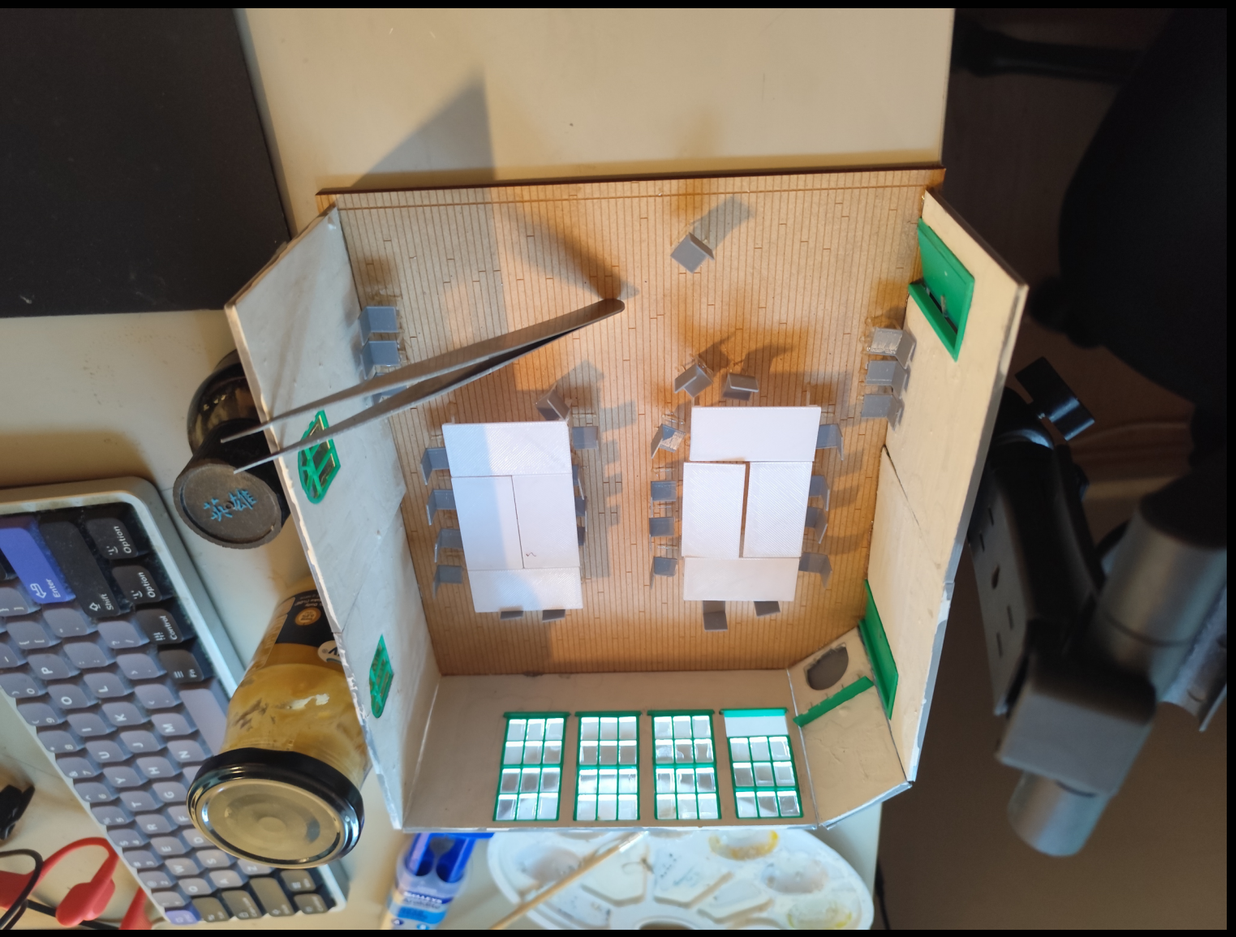
## Assemble

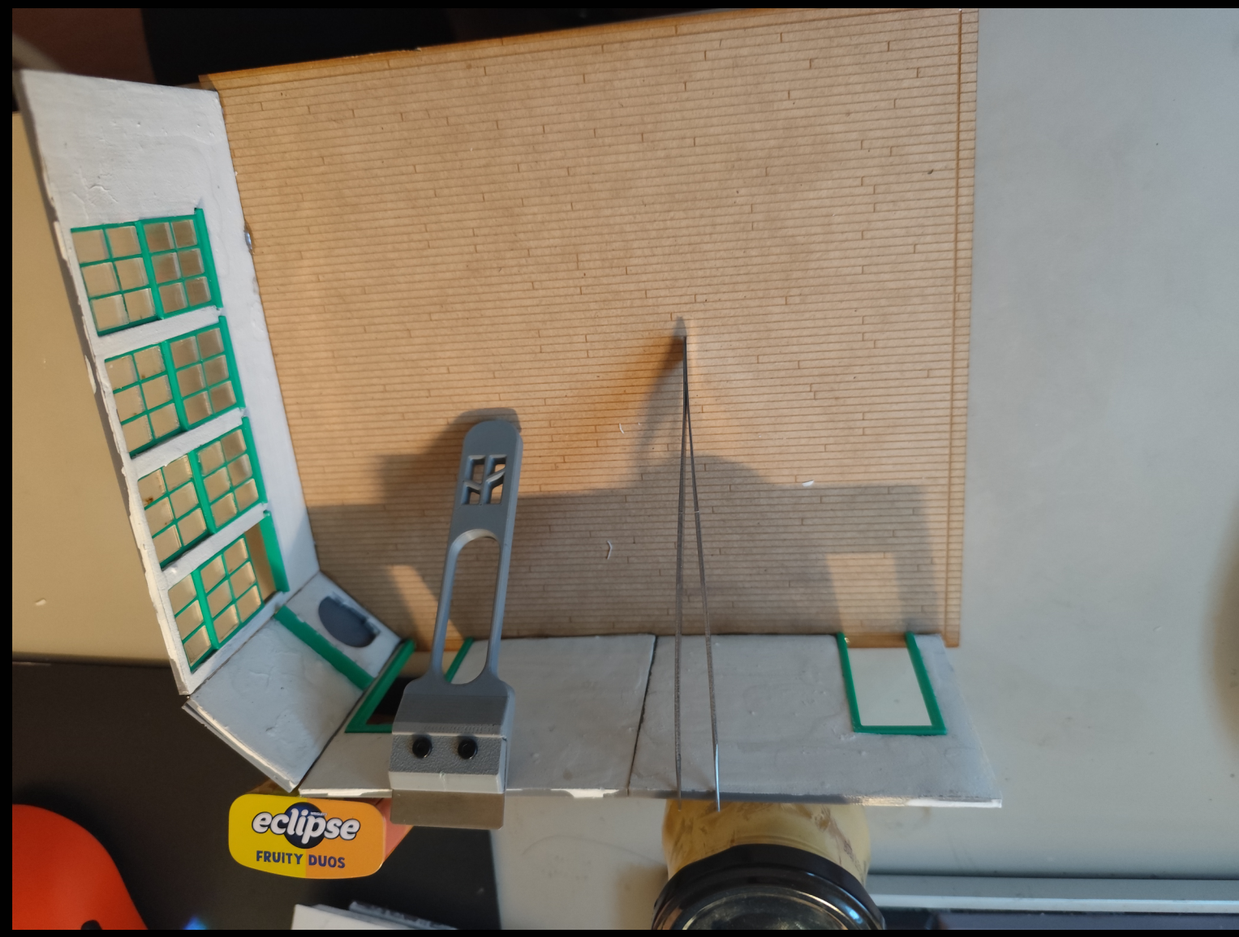
The room was assembled according to the manual, and for better display, a wall without any elements was not added to allow the audience to better observe the interior.

During the final assembly, multiple videos were filmed and combined to showcase the process.

https://youtu.be/05SeOXvB8_k






eclipse
FRUITY DUOS